Does Listing Language Add Value Beyond Location? Deconfounding Text and Geography in Automated Home Valuation

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Abstract

Automated valuation models increasingly price homes by reading the prose of a listing alongside its structured attributes, and the language reliably improves their predictions. Whether that improvement reflects genuine property information or merely the neighborhood an agent's description encodes has not been tested, and the distinction matters: a model that prices laundered geography offers nothing beyond location and inherits its biases, while one that prices semantic content adds real information. We treat the leading direction of a listing embedding as a treatment and estimate its effect on log price with a partially linear double machine learning score, holding the structured attributes and the parcel's location fixed. Across twelve metropolitan markets the price effect of listing language is almost unmoved by spatial adjustment in eleven; the exception is New York, whose text is the most geographic of the twelve. The effect also holds on recorded sale prices in the four markets where transactions are public, though it attenuates, since part of the asking-price association is the agent's own pricing of the language. The estimates are associational, and we bound their fragility with a sensitivity analysis. In most markets, a listing's language is priced for its content, not its geography.

Keywords: Spatial confounding, Concept erasure, Double machine learning, Generated regressors, Text as data, Real estate valuation

1 Introduction

Automated valuation models increasingly read the natural-language description of a property alongside its structured attributes, and the text reliably improves their predictions; Shen and Ross [2021], Baur et al. [2023a], and Lorenz et al. [2023] all report sizable reductions in pricing error once listing-text embeddings are added on top of location and the usual structured controls. The reading attached to that improvement is the optimistic one, that listing language carries information about a property its tabular attributes miss, and on its face it is plausible, since a good description does say things about condition, light, and finish that no bedroom count records. What predictive accuracy cannot do is tell that interpretation apart from a deflationary one. Property descriptions are written by agents who know exactly where a listing sits, so any embedding of those descriptions absorbs the neighbourhood whether the encoder intends it to or not, and a model that prices the embedding may be doing little more than relabelling location as semantic content. Because the two stories make identical predictions, the gain that motivates text-augmented valuation is by construction silent on whether the language adds anything at all once geography is accounted for, and it is that silence the paper sets out to break.

Breaking that silence means asking how much of the association between a listing's language and its price is owed to geography rather than to anything the language contributes on its own. The question is an instance of spatial confounding, the difficulty in which an exposure of interest and an unmeasured driver of the outcome share a geographic pattern, so that ordinary regression hands the confounder's effect to the exposure [Paciorek, 2010, Hodges and Reich, 2010]. Our test holds the text fixed and moves geography on and off the controls. We summarise a description by the leading direction of its embedding, estimate that direction's effect on log price with a partially linear Double Machine Learning score [Robinson, 1988, Chernozhukov et al., 2018], and read the change in the effect when a flexible location basis is added to the controls, the contrast a Gelbach decomposition makes precise. That the representation encodes location at all is not in doubt; we confirm it with a concept-erasure diagnostic, developed in Appendix D, that recovers the coordinate from the embedding and then projects it out. The question the paper turns on is the separate and more consequential one of whether that encoded location actually drives the price effect. Running the test across twelve metropolitan markets lets the answer be read as a distribution rather than a single number, since the embedding geometry, the structured controls, and the listing density that decide how much room the text has once location is accounted for all differ from one market to the next.

Applied across twelve metropolitan areas using 69,173 deduplicated Redfin listings, the answer is more reassuring than the predictive-fit worry would suggest. A held-out probe recovers the latitude-longitude coordinate from the embedding at an R^2 approaching one half, so the representation plainly carries location. And yet holding the text direction fixed and folding a flexible location basis into the controls leaves its per-standard-deviation effect on log price almost exactly where it began in eleven of the twelve markets. The one market in which the adjustment bites, New York, is the market whose text direction is itself the most geographic, so the exception illuminates the mechanism rather than embarrassing it. The surviving effect does not depend on the encoder: a Doc2Vec uniqueness channel and a sentence-BERT principal component move almost in lockstep across markets, two encodings of one signal rather than independent confirmations of it. In economic terms the effect is sizable rather than merely detectable: a one-standard-deviation move along the text direction is worth between roughly eight and seventeen percent of a home's price, depending on the encoder, several times the marginal value of an additional bedroom. One inferential subtlety needs care. The leading principal component is estimated on the same sample that estimates the price effect, which makes it a generated regressor in the sense of Pagan [1984], and its analytic interval undercovers; we document this in a calibration study and repair it with a bootstrap over the full pipeline (Appendix G).

None of this is a clean causal elasticity, and we say so throughout. The estimates are associational, and a sensitivity analysis in the style of Cinelli and Hazlett [2020] reports how weak an unobserved confounder would have to be to overturn each market's result, a threshold that comes out uncomfortably modest in several of them. Because the outcome is the asking price an agent sets together with the description, we ask finally whether the effect holds on the price a property actually fetches. In the four markets that publish recorded transactions it replicates on sale prices, though it attenuates, which is what we would expect when part of the asking-price association is the agent's own pricing of the language. The answer the paper returns to the measurement question the text-valuation literature left open is therefore this: in most metropolitan markets a listing's language is priced for its content and not merely for the geography it encodes, with New York the documented exception. We release the code, the derived embeddings, the public-record sale prices, and an independent computational verification of every analytic claim (Appendix F); the listing text and asking prices are scraped under a commercial platform's terms and are not redistributed, but the collection code that reconstructs them is.

The paper proceeds in the order a skeptical reader would want. Section 2 places the question among the literatures it draws on. Section 3 sets out the estimand and the partialling-out estimator, and Section 4 describes how the twelve-market corpus was assembled and audited for the duplication that scraped listings invite. The evidence occupies Section 5, which estimates the text effect under each encoder, checks how much location the representation encodes, holds the text fixed and moves geography on and off the controls to isolate the share that location explains, bounds the surviving effect with a sensitivity analysis, and confirms it on recorded sale prices; an auxiliary rewriting intervention and a cross-encoder comparison are reported alongside. The concept-erasure implementation and the formal treatment of the generated-regressor problem are developed in the appendices. Section 6 draws the line, for practice, between what a predictive-only evaluation of text-augmented valuation can and cannot support.

2 Related Work

2.1 Listing text in automated valuation, and the question it raises

The use of natural language in property valuation predates the recent enthusiasm for embeddings, and the empirical claim that listing text improves prediction is by now well established. Nowak and Smith [2017] show that the free-text remarks attached to a multiple-listing record reduce pricing error substantially once they are encoded, and Nowak et al. [2021] document that the vocabulary doing the predicting shifts from one submarket to the next. A particular channel is formalised by Shen and Ross [2021], who measure the textual uniqueness of a description relative to its neighbours and report a uniqueness premium on the order of ten to fifteen percent per standard deviation in a single metropolitan market, while Baur et al. [2023a] show that folding a sentence-embedding representation into a structured valuation model lowers its error by as much as a sixth. With few exceptions the literature reads gains of this kind as evidence that language carries information about a property which its tabular attributes do not.

What that reading passes over is that an agent writes a description already knowing exactly where the property sits, so the language is saturated with location before any encoder touches it, and a representation learned from such text will carry the neighbourhood whether or not it carries anything more. Because a predictive metric cannot tell a model that prices genuine semantic content apart from one that prices laundered geography, the very gains that motivate text-augmented valuation are silent on which of the two is operating, and the question of how much of the apparent text effect is real has gone largely untested in a causal register even as the predictive work has matured. We take up that question directly, and the apparatus it calls for sits at the meeting point of three literatures that have seldom been brought into the same room, namely the treatment of text as a causal variable, the spatial-statistics account of confounding by location, and the concept-erasure methods developed in natural language processing for removing a named attribute from a representation. The same machinery bears on the fairness and auditability concerns that motivate much of the recent scrutiny of automated valuation, since Zhu et al. [2024] and Guo [2025] document residual valuation gaps that survive architectural improvement and a text feature re-encoding neighbourhood geography re-encodes whatever that geography is correlated with, but the object of this paper is the prior measurement question of whether the text effect is genuine at all rather than the disparities a confirmed effect would carry.

2.2 From Hedonic Models to Multimodal Valuation

The idea that property values can be decomposed into contributions from observable characteristics dates to the hedonic pricing framework [Rosen, 1974, Sheppard, 1999]. Automated valuation models built on this foundation through progressively richer statistical machinery: generalized additive models [Gloudemans, 1999], spatial autoregressive specifications that account for price correlation among neighboring properties [Pace and Barry, 1998, Can, 1992], and kriging-based approaches that interpolate unobserved spatial surfaces [Bourassa et al., 2010]. Each generation recognized more clearly that spatial dependence is not a nuisance to be corrected but a first-order feature of housing markets.

Machine learning accelerated this trajectory. Ensemble methods outperform linear hedonic models by capturing non-linear feature interactions automatically [Park and Bae, 2015, Antipov and Pokryshevskaya, 2012], and deep learning extended the input space beyond tabular features to images [Poursaeed et al., 2018, Law et al., 2019] and text [Shen and Ross, 2021, Baur et al., 2023a]. Visual models trained on property photographs or street-view imagery can estimate luxury level and architectural style [Poursaeed et al., 2018, Lindenthal and Johnson, 2021], while textual models extract semantic features from listing descriptions that correlate with sale prices [Nowak and Smith, 2017, Ahmed and Moustafa, 2016]. Multimodal systems fuse these modalities together, reporting performance gains attributed to complementary information [Bin et al., 2020, Kang et al., 2021, Kostić and Jevremović, 2020]. Graph neural networks push the idea further by propagating information between spatially proximate properties, so that each node's representation absorbs its neighbors' features [Das et al., 2021, Lin et al., 2021, Peng et al., 2022, Li et al., 2018, Huang et al., 2018].

A persistent concern runs through this literature, raised early but seldom addressed. The most predictive words in property descriptions vary across geographic submarkets [Nowak and Smith, 2017, Nowak et al., 2021], visual features capture neighborhood-level rather than property-level variation [Law et al., 2019], architectural style clusters geographically [Lindenthal and Johnson, 2021], and GNN message passing, by design, propagates whatever signal is shared among neighbors, including location-induced price similarity. If text, images, and graph features all encode geography, then fusing them creates the appearance of complementary signal from what is really redundant spatial encoding through different channels.

2.3 The Semantic Interpretation and Its Limits

The commercial and research appeal of text-based valuation rests on a specific claim: that property descriptions carry information about value that structured features miss. A one-standard-deviation increase in textual "uniqueness" is reported to raise predicted price by 10–15% [Shen and Ross, 2021]. Including text embeddings is reported to reduce valuation error by up to 17% [Baur et al., 2023a]. Hierarchical graph representations that embed similar-value properties near one another in latent space appear to learn price-relevant structure [Zhang et al., 2021]. UMAP projections of these embeddings produce visually appealing clusters that seem to correspond to market segments [Baur et al., 2023a].

Every one of these findings is also predicted by a model in which text carries zero independent information. If agents in affluent neighborhoods write "luxury finishes" and "steps from fine dining" while agents in family-oriented suburbs write "great school district" and "quiet cul-de-sac," the embeddings will cluster by price segment, SHAP will rank descriptive terms as important [Lorenz et al., 2023, Stang et al., 2023], and adding text to a location-only model will improve R^2, even though the entire gain comes from spatially confounded language rather than property-specific content. The assumption that generic quality terms like "spacious" or "renovated" provide location-independent signal [Lorenz et al., 2023] ignores that "spacious" is more frequent in suburban markets and "renovated" more frequent in gentrifying urban neighborhoods. Nowak et al. [2021] find exactly this: the predictive vocabulary shifts across submarkets, suggesting that language mirrors geography rather than transcending it. Wan and Lindenthal [2023] raise the same concern for visual features, proposing system-level tests to determine whether models respond to building attributes or to irrelevant context.

The distinction between correlation and causation is not academic here. If text embeddings genuinely encode value-relevant property attributes independent of location, then NLP-enhanced valuation models offer real information gains. If they primarily encode geography through linguistic patterns, then these models are opaque spatial models and their accuracy gains are expected, not evidence of new signal. Yet across the studies reviewed above, none tests the causal claim. The tools to do so exist in causal inference and spatial statistics, and the confounding structure is transparent, but the question has not been asked. This paper asks it.

2.4 Spatial Confounding and Attribute Leakage in Learned Representations

The concern that text may encode geography rather than semantics connects two literatures that have developed largely in parallel.

In spatial statistics and epidemiology, the problem is well characterized: when an exposure of interest varies spatially and shares its spatial pattern with unmeasured confounders, standard regression attributes the confounders' effects to the exposure [Paciorek, 2010]. The severity depends on the spatial scale: fine-grained exposures are most vulnerable to fine-grained confounders. Because that scale is rarely known in advance, Keller and Szpiro [2020] cast the flexibility of the adjustment as a model-selection problem, fixing it by the Euclidean distance at which spatial variation is removed rather than by an opaque degrees-of-freedom count. Gilbert et al. [2021] set out the conditions under which spatial coordinates can stand in for an unmeasured spatial confounder: the confounder must be a smooth function of location, and the exposure must retain variation that is not itself spatial. The second condition is the one a learned treatment can fail, and our erasure diagnostic shows the text direction to satisfy it; Gilbert et al. [2024] study the consistency of the resulting estimators. Methods to address this include restricted spatial regression, which orthogonalizes spatial random effects against fixed effects of interest [Reich et al., 2021], and sensitivity frameworks that quantify the confounder strength needed to overturn a finding [Cinelli and Hazlett, 2020]. A counterintuitive result from this literature is that adding spatially-correlated error terms to a model can worsen, not improve, inference about fixed effects when spatial confounding is present [Hodges and Reich, 2010]. Spatial econometrics addresses the same challenge through spatial lag and error models [Anselin, 1988, LeSage and Pace, 2009, Elhorst, 2014], though identification of spatial dependence parameters remains difficult to separate from omitted-variable bias [Gibbons and Overman, 2014].

In NLP, a parallel line of work has shown that text embeddings absorb far more than semantic content from their training corpora. Word embeddings encode human-like gender and racial biases [Bolukbasi et al., 2016], replicate implicit association test results [Caliskan et al., 2017], and retain these biases even after standard debiasing [Gonen and Goldberg, 2019]. Language use varies systematically by geography: user location can be predicted from tweets with high accuracy [Rahimi et al., 2018], and lexical innovations diffuse through social networks along spatial paths [Eisenstein et al., 2014]. Property descriptions are an extreme case of this phenomenon because agents write with complete knowledge of the property's location, neighborhood, and market context. The language is not incidentally spatial; it is intentionally so.

Machine learning has begun to develop tools for this class of problem. Causal representation learning seeks representations that reflect causal structure rather than spurious correlation [Schölkopf et al., 2021, Peters et al., 2017]. Counterfactual fairness formalizes the requirement that predictions be invariant to interventions on protected attributes [Kusner et al., 2017, Kilbertus et al., 2017]. Domain-adversarial networks learn confounder-invariant representations through gradient reversal [Ganin et al., 2016], and variational approaches attempt to disentangle causal from spurious factors [Louizos et al., 2017, Moyer et al., 2018]. The invariance of causal mechanisms across environments, in contrast to the instability of correlations, motivates transfer-based identification [Rojas-Carulla et al., 2018]. Causal discovery algorithms infer graph structure from data using conditional independence tests [Spirtes et al., 2000], score-based search [Chickering, 2002], or functional asymmetries between causal directions [Shimizu et al., 2006, Peters et al., 2016]. Double Machine Learning [Chernozhukov et al., 2018] provides a semiparametric framework for estimating causal effects with machine-learned nuisance parameters at $\sqrt{n}$ rates.

These threads meet in the setting we study, and they have begun to meet elsewhere as well, which is worth stating precisely so that what is new here is not overstated. The idea of adapting a text representation for causal rather than predictive ends is developed by Veitch et al. [2020], whose embeddings are built to retain the confounding-relevant subspace, and the inverse operation of stripping a concept from a document embedding by linear erasure is carried out for similarity and clustering by Fan et al. [2025], while Feldman et al. [2026] treat latent textual features as a treatment whose covariate content has to be residualised away. What this paper adds is therefore neither the erasure operator nor the recognition that a text feature can conflate treatment with confounder, both of which are by now established, but the application of that apparatus to an economic estimand, the price effect of a listing's language, carried out with a design that separates the geography a representation merely encodes from the geography that actually moves the outcome and with inference suited to an embedding-valued treatment. Property valuation is an unusually clean place to attempt this, since location drives both the text and the price, the outcome is measured precisely, and the practical stakes are immediate, and we run the analysis across twelve metropolitan markets rather than the single market the hedonic-text literature has usually examined, which lets the answer be read as a distribution and lets the one market where location does bite be identified rather than assumed away.

3 Methodology

3.1 Setup and the conflation it invites

Write $\mathcal{D} = \{(x_i, t_i, \ell_i, y_i)\}_{i=1}^n$ for a market of n listings, where $x_i \in \mathbb{R}^d$ collects the structured property attributes that hedonic models have always used, the bedroom and bathroom counts, the living area, the lot size, and the year built; $\ell_i \in \mathbb{R}^2$ is the latitude-longitude coordinate of the parcel; y_i is the log listing price; and t_i is a representation of the listing's free-text description, either a scalar summary or a high-dimensional embedding depending on the channel. A text-augmented valuation model learns some $\hat{y}_i = f(x_i, t_i, \ell_i)$, and the empirical regularity that motivates this literature is that adding t_i to the feature set raises predictive accuracy beyond what x_i and ℓ_i deliver on their own.

The difficulty is that an accuracy gain cannot say what kind of information the text carried. An agent who writes "sun-drenched and steps from the park" is describing a property, but the same phrase is also a near-deterministic signal of which block the property sits on, because language and location are produced together by an author who observes both. A predictive model that prices genuine semantic content and a predictive model that prices laundered geography make identical out-of-sample predictions, so the gain on its own leaves an analyst unable to tell which of the two has been bought. Our object of interest is the part of the text-price relationship that is not the spatial information the text carries, and the methods that follow are built to isolate it and then to ask, for each metropolitan market in turn, how much of the apparent text signal that isolation leaves standing.

This conflation has a precise name in the literature on text as a treatment, where it appears as the aliasing of one measured latent feature of a document with the unmeasured features that the same prose inevitably carries, and Fong and Grimmer [2023] show that even a randomly assigned text fails to identify the effect of a single latent feature unless the others it travels with are either independent of it or irrelevant to price. We do not impose that condition, so the quantity we isolate is the price effect of the leading text direction marginalizing over whatever else the language encodes, an associational object rather than the effect of a cleanly separated semantic attribute, and we are explicit about that reading throughout. A second lesson from this literature bears on estimation rather than interpretation, since Egami et al. [2022] establish that when the treatment representation is learned from the documents it must be formed on data held apart from the outcome, which is the role the out-of-fold construction of the treatment direction described below plays here.

3.2 A structural account of the confounding

The data-generating process we have in mind makes the confounding explicit rather than assuming it away. Location is exogenous and drives everything downstream: the structured attributes x, a vector c of contextual neighborhood features such as demographics, amenities, and recent crime, the text t, and the price y. The text in turn is written with knowledge of location, structure, and context, so $t = f_t(\ell, x, c, U_t)$, and the question the paper exists to answer is whether t enters the structural equation for y at all once ℓ, x, and c are accounted for, or whether the entire observed association between text and price runs through the common dependence of both on location.

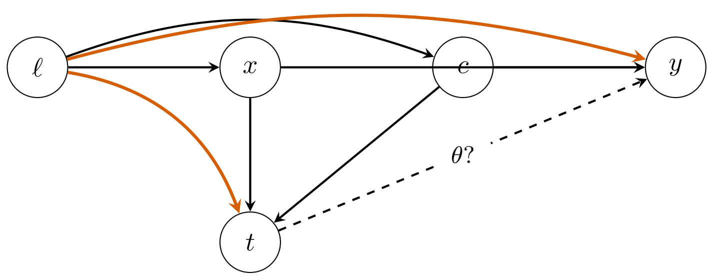

Figure 1: The structure the paper adjudicates. Location ℓ is exogenous and feeds the property attributes x, the contextual features c, the listing text t, and the price y. The two highlighted edges $\ell \to t$ and $\ell \to y$ form the spatial-confounding backdoor, the reason text and price are associated in part simply because both depend on where the property sits. The dashed edge labelled θ is the direct effect of text on price, the quantity we estimate, and the empirical question is how much of the text-price association is that direct edge rather than the backdoor through ℓ. Conditioning on $\{\ell, x, c\}$ blocks the backdoor and identifies θ; whether θ is zero or positive is what the twelve markets decide.

Two readings of this diagram bracket the possibilities. Under one, the text effect on price is zero net of location and the structured covariates, every observed text-price correlation is geography wearing a semantic costume, and the apparent predictive value of listing language is spatial confounding mediated through words. Under the other, listing text carries property-specific content that the structured covariates miss, so t does enter the equation for y and the text-price relationship is at least partly a genuine semantic effect. The two readings disagree about a quantity that is identified once the backdoor set $\{\ell, x, c\}$ is conditioned on, since that set blocks every path from t to y that does not pass through the hypothesized direct edge, a backdoor argument made precise in Appendices A and B, and the rest of this section develops the estimator, the encodings, and the deconfounding tests through which the twelve markets adjudicate between them.

3.3 The partially-linear estimand and its estimation

We summarize the listing text by a scalar treatment T, defer to Section 3.4 how that scalar is constructed, and pose the partially-linear model

$$Y = \theta T + g(\ell, x, c) + \varepsilon, \qquad T = m(\ell, x, c) + \nu, \tag{1}$$

in which θ is the per-standard-deviation effect of the text direction on log price and the nuisance functions g and m are left unrestricted. The Robinson partialling-out estimator [Robinson, 1988] recovers θ from the residual-on-residual regression of $\tilde{Y} = Y - \hat{g}$ on $\tilde{T} = T - \hat{m}$. We fit the nuisances $\hat{g}$ and $\hat{m}$ by cross-fitted machine learning in the double machine learning framework of Chernozhukov et al. [2018], using gradient boosting for the embedding-based channels and ridge regression for the Shen-Ross uniqueness channel in keeping with its published hedonic protocol. Each market is split into $K = 5$ folds, and each fold's residuals are predicted from models trained on the complement, so that the nuisance estimates and the score are never formed on the same observations; Appendix J consolidates the learner, fold, and bootstrap settings. The resulting estimator is Neyman-orthogonal and root-n consistent whenever the nuisance fits converge faster than $n^{-1/4}$, and its influence-function variance gives the standard error we report by default,

$$\hat{\theta} = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}, \qquad \hat{\mathrm{SE}}_\theta = \sqrt{\frac{1}{n^2} \sum_i \frac{(\tilde{Y}_i - \hat{\theta}\,\tilde{T}_i)^2\,\tilde{T}_i^2}{\left(\frac{1}{n}\sum_j \tilde{T}_j^2\right)^2}}. \tag{2}$$

One feature of the embedding channel complicates the inference and we flag it here because it shapes how the per-market intervals should be read. When the treatment is the leading principal component of the embedding, that direction is itself estimated on the same sample that estimates the price effect, so T is a generated regressor in the sense of Pagan [1984], the machine-learning-generated case treated by Battaglia et al. [2024], and the orthogonal-score variance in Equation (2) does not absorb the extra sampling variability the estimation of the direction introduces. The interval is correspondingly anticonservative at the few thousand listings a single market affords, mildly so near the null and more seriously when the true effect is large, a pattern we document in a calibration study and repair with an out-of-fold construction of the treatment direction, the sample-splitting that Egami et al. [2022] prescribe when the treatment is learned from the documents, in Appendix G. The point estimates are approximately unbiased throughout; it is the nominal coverage of the analytic interval, not the location of the estimate, that the calibration study qualifies.

3.4 Two encodings of the text treatment

Because no single reduction of a paragraph to a number is obviously the right one, we read the listing text through two encodings with different inductive biases and treat their agreement as a robustness check rather than as independent corroboration. The first follows the hedonic specification of Shen and Ross [2021], who summarize a description by a Doc2Vec uniqueness score that measures how far a listing's language departs from the corpus norm, and we enter that scalar into Equation (1) as the treatment. The second follows Baur et al. [2023a], who embed each description with a pretrained sentence transformer and read the leading principal component of the embedding as the text signal; we standardize that component and use it as the treatment in a separate partialling-out fit. The sign of a principal component is not identified, since a component and its negation span the same axis, so wherever the Baur channel appears we orient the common axis to make the pooled estimate non-negative and disclose the orientation as a convention rather than a finding, a convention that cannot manufacture agreement because the cross-market correlation between channels is invariant in magnitude under a sign flip of either one.

3.5 What the embedding encodes: erasure as a diagnostic

The sentence-transformer embedding that underlies the Baur channel was pretrained on generic text and has no reason to keep semantic content about a property separate from the geographic information that correlates with its neighborhood, so before asking whether the price effect is confounded we ask the prior question of how much location the representation carries at all. We answer it by erasing the coordinate concept from the embedding and measuring what an erasure leaves recoverable. The closed-form concept erasure of Belrose et al. [2023] removes the linear subspace that the mean-conditional-covariance estimator associates with the latitude-longitude pair and comes with an exact-guardedness guarantee that no linear predictor can recover the coordinate from the residual better than a constant baseline, and the iterative nonlinear procedure of Iskander et al. [2023] extends the attack to the directions a linear probe cannot see by alternating between fitting a nonlinear probe of the coordinate and projecting out where that probe concentrates its predictive mass. The erasure is a diagnostic of what the representation contains and not a modification of any per-market estimate, since the Baur effects are computed on the unprojected embedding in keeping with the published protocol; its role is to bound how much location a linear or nonlinear function of the text could carry, which the decomposition below then connects to how much location actually moves the price.

3.6 When a probe fails, what has been erased?

The erasure diagnostic of the previous subsection rests on probing, and probing carries a subtlety that bears on how its verdicts should be read. The natural way to check that a concept has left a representation is to train a probe of the concept on the cleaned representation and confirm that it does no better than chance, yet a probe that fails establishes only that this particular probe, at this capacity, cannot recover the concept, and not that the representation has shed it, since a fresh classifier of higher capacity fit after the fact can recover what a weaker or adversarially pressured probe could not. This is the reason we lean on closed-form erasure, whose guardedness guarantee holds against every linear probe at once rather than against the one we happened to train, and the reason we read the residual probe R^2 that survives the iterative erasure of Section 5.3 as a lower bound on the location a nonlinear function could still recover rather than as a measurement of zero. The same subtlety is a warning about the obvious alternative to closed-form erasure, an adversarially trained encoder whose discriminator gradient reversal drives to chance, because a live discriminator at chance is the weakest possible evidence of erasure: it certifies only that the encoder defeated the one probe it was trained against, and a fresh post-hoc probe of greater capacity may still read the concept off the same representation. We make the point precise as a $\mathcal{V}$-information [Xu et al., 2020] statement connected to the Barber–Agakov variational lower bound used throughout the probing literature [Pimentel et al., 2020].

For any class $\mathcal{F}$ of probabilistic classifiers $q : \mathcal{Z} \to [0, 1]$, define the variational lower bound on $I(Z; C)$:

$$V_{\mathcal{F}}(Z; C) \;:=\; \sup_{q \in \mathcal{F}} \big[H(C) - \mathbb{E}_{(Z,C)}[-\log q(C \mid Z)]\big]. \tag{3}$$

The supremum is achieved at $q^\star = p(C \mid Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior; otherwise $V_{\mathcal{F}} < I(Z; C)$ strictly.

Proposition 1 (Frozen-probe diagnostic). *Let $E_\phi : \mathcal{X} \to \mathcal{Z}$ be an encoder, Φ the discriminator class trained jointly via gradient reversal, and Φ' a probe class with $\Phi \subseteq \Phi'$ trained post-hoc on the frozen encoder. Then:*

(a) $0 \le V_\Phi(Z; C) \le V_{\Phi'}(Z; C) \le I(Z; C)$, *with the rightmost inequality strict whenever Φ' omits the Bayes-optimal posterior.*

(b) The empirical Barber–Agakov lower bound under Φ equals the gradient-reversal discriminator's converged value: $\hat{V}_\Phi(Z_{\hat{\phi}}; C) = H(C) - \hat{L}_\Phi$. Therefore "the live discriminator achieves chance" is equivalent to $\hat{V}_\Phi \le \epsilon$. Under uniform-convergence conditions on Φ', the post-hoc plug-in $\hat{V}^{post}_{\Phi'}(Z_{\hat{\phi}}; C) \to V_{\Phi'}(Z_{\hat{\phi}}; C)$ in probability, and hence $\hat{V}^{post}_{\Phi'} - \hat{V}_\Phi$ is a consistent lower-bound estimator of the residual mutual information $I(Z_{\hat{\phi}}; C) - V_\Phi(Z_{\hat{\phi}}; C) \ge 0$.

(c) There exist a data distribution over (X, C), a downstream label Y with $I(X; Y) > 0$, class pairs $\Phi \subsetneq \Phi'$, and a joint-game weight $\alpha > 0$ such that the joint adversarial-prediction loss

$$\mathcal{L}(\phi, g, h) \;=\; \alpha \cdot \mathcal{L}_{task}(g(E_\phi(X)), Y) \;-\; \mathcal{L}_{disc}(h(E_\phi(X)), C)$$

admits a saddle $(\phi^\star, g^\star, h^\star)$ with $V_\Phi(Z_{\phi^\star}; C) = 0$ but $V_{\Phi'}(Z_{\phi^\star}; C) \ge H(C) - \delta$ for arbitrary $\delta > 0$.

Statement (a) is folklore in the variational mutual-information literature, appearing as Proposition 1 of Xu et al. [2020] in the language of $\mathcal{V}$-information and as the central observation of Pimentel et al. [2020] in the language of probe capacity, and we restate it for self-containedness. Statements (b) and (c) are our contribution. Part (b) ties the bound to the gradient-reversal objective and recasts the post-hoc adversary used as a heuristic by Moyer et al. [2018] and Elazar and Goldberg [2018] as a consistent statistical estimator of residual leakage, which is what licenses reading the surviving probe R^2 as a number rather than a guess. Part (c) establishes that the joint adversarial-prediction game admits saddles at which the live discriminator achieves the optimum of its own minimax game while essentially all of the concept remains recoverable to a richer probe, so an adversarial erasure can certify success and be wrong. The existential construction mirrors the style of Ravfogel et al. [2022] Proposition 3.2 and Belrose et al. [2023] Theorem 4.3, and proofs together with three synthetic numerical experiments verifying each part are deferred to Appendix E.

3.7 Holding the text fixed: a covariate decomposition

Showing that the embedding encodes geography is not the same as showing that the price effect is confounded by it, because a representation can carry a confounder along directions that have nothing to do with the one on which the treatment acts. The test that speaks to confounding directly holds the text treatment fixed and moves geography across the conditioning set, in the manner of the order-invariant covariate decomposition of Gelbach [2016]. We estimate the text effect twice on the same fixed treatment, once with the location coordinates held out of the controls and once with a flexible spatial basis, the coordinates together with their squares and their interaction, folded in, and read the difference between the two estimates as the share of the text effect that explicit geography accounts for. A small difference means the text-price effect is not the spatial information re-expressed, and the decomposition is informative rather than mechanical precisely when the text direction is shown by the erasure diagnostic to carry geography, since a direction that encodes location yet whose price effect is unmoved by conditioning on location is the signature of a genuine semantic signal. This contrast is the paper's headline deconfounding test, and Section 5.4 reports it for every market.

3.8 An intervention check and a sensitivity bound

Two further instruments probe the same question from angles with different failure modes. A counterfactual rewriting intervention uses an open-weight language model to edit each description under a minimal-pair protocol, producing one rewrite that strips stylistic content while holding the model's inferred submarket fixed and a second that also moves the targeted submarket, and the price differentials these edits imply isolate a within-listing style contrast and a location-inference contrast whose biases are not shared with the cross-listing geometry of a fixed encoder. Because none of these designs identifies a structural effect by assumption, we close with the sensitivity calculus of Cinelli and Hazlett [2020], reporting for each market the robustness value, the strength an unobserved confounder would need on both the treatment and the outcome to overturn the estimate, benchmarked against the strength of the spatial covariates we do observe. The twelve per-market estimates are finally combined through a random-effects meta-analysis with the Paule-Mandel variance estimator and Hartung-Knapp-Sidik-Jonkman small-sample intervals [Paule and Mandel, 1982, IntHout et al., 2014], and a moderator scan asks, with multiplicity and leverage diagnostics, whether any metropolitan covariate explains the heterogeneity the markets display.

4 Dataset Construction

4.1 A twelve-market corpus of listings

The corpus is built from sold residential listings scraped from a major online real estate platform across twelve metropolitan markets that span the price and geographic range of the United States housing stock, from San Francisco and New York at the expensive coastal end through Boston, Seattle, Washington, Denver, and Portland in the middle to Chicago, Atlanta, Phoenix, Dallas, and Philadelphia toward the more affordable interior and south. Each listing carries the free-text description an agent wrote, the asking price the platform reports for the property at the time of the crawl, the structured attributes a hedonic model expects, the bedroom and bathroom counts, the living area, the lot size, the year built, and the property type, and a property-level latitude and longitude rather than a coarse centroid. Two features of this outcome we are explicit about. The price is the listing's asking price rather than a realized transaction, and it is set by the same agent who writes the description and at the same moment. Part of any text-price association is therefore the agent's own pricing of the language rather than the market's, a simultaneity that no spatial adjustment removes. It limits the causal reading of the estimates along an axis orthogonal to the spatial-confounding question this paper takes up. Because the corpus is a single cross-sectional crawl, the outcome itself carries no calendar-time variation that pooling could confound; the property's most recent prior-sale date, which we recover from the listing page for every property and release with the data, enters only as a covariate and is in any case essentially uncorrelated with the asking price within each market, the absolute correlation no larger than 0.20 and the R^2 below 0.04.

Table 1: The twelve-market corpus. *Listings* is the per-market analysis sample after deduplication to one row per listing and removal of rows missing a required field; the markets together contribute 69,173 listings. Median sale price and median description length are computed on the analysis sample. The panel ranges over more than an order of magnitude in size and roughly fivefold in price.

Market	Listings	Median price	Median words
New York	15,227	\$975,000	183
Dallas	8,008	\$439,000	161
Phoenix	7,086	\$514,990	117
Philadelphia	7,030	\$255,000	134
Atlanta	6,134	\$384,000	178
Chicago	5,576	\$374,900	129
Denver	5,280	\$525,000	207
Portland	4,337	\$559,000	166
Washington	4,015	\$599,000	187
Seattle	2,884	\$750,000	152
Boston	2,610	\$999,000	137
San Francisco	986	\$1,295,000	174
Total	69,173		

4.2 Deduplication, outliers, and a robust winsorizer

The scraper issued repeated requests against the same listing during collection, so the raw rows resolve to a smaller set of distinct listings once they are deduplicated on the listing identifier, and every number in the paper is computed on the one-row-per-listing analysis sample to keep sibling copies of a listing out of the training folds of any cross-fitted procedure. We drop transactions outside a plausible price band to remove data-entry errors and non-arm's-length transfers, and we guard the structured covariates with a per-column robust winsorizer that clips each feature at its median plus or minus five median absolute deviations and imputes the rare non-finite entry at the median. The winsorizer matters more than its routine description suggests. A single parcel-join artifact, a year-built value tens of standard deviations from any plausible construction date, was enough to make the partialling-out estimator in one market ill-conditioned and to drive its coefficient to an impossible magnitude with a deceptively small standard error, and clipping the genuine outliers without touching clean data restored that market to the range of the others while leaving every well-behaved market unchanged. The clip is applied uniformly across the twelve markets and is the only manipulation of the structured covariates we perform.

4.3 Confounders

The conditioning set is assembled to make the backdoor adjustment of Section 3 credible at the resolution where language and price are both determined. Alongside the structured property attributes, we attach the property's location through its coordinates and its ZIP code, and we join neighborhood context by spatial position rather than by a coarse identifier. Census demographics come from the American Community Survey five-year estimates at the block-group level, the finest geography the survey supports, and contribute median household income, educational attainment, racial composition, median home value, and median gross rent for the block group containing each listing. Recent crime is counted within a five-hundred-meter radius of each property from municipal incident records, separated into violent, property, and quality-of-life categories through a cross-city offense crosswalk. Local amenities are counted in the same radius from an open geographic database, covering food and retail, services, recreation, transit, and education. The assembled set runs to roughly three dozen controls per market, and the spatial block within it, the coordinates together with the ZIP indicators, is the group the sensitivity analysis of Section 5.5 singles out because it is the one most directly implicated in the text-as-spatial-proxy concern. Appendix C documents the scrape, the geocoding, and the census, crime, and amenity joins.

4.4 Text preprocessing and the two encodings

Before encoding, each description is stripped of markup and boilerplate, lowercased, and cleared of explicit street addresses, which are replaced with anonymization tokens so that a model cannot recover location by reading an address out of the text rather than out of the language. The cleaned text then feeds the two encodings that Section 3.4 defines. The Doc2Vec uniqueness score that drives the Shen-Ross channel measures how far a listing's language sits from its market's corpus norm, and the sentence-transformer embedding that drives the Baur channel is the 768-dimensional representation of `all-mpnet-base-v2` [Reimers and Gurevych, 2019], from which we take the standardized leading principal component as the text treatment. We recompute the full analysis on a second, smaller transformer to confirm that the conclusions are not an artifact of one encoder's geometry, and the deconfounding and erasure diagnostics of Sections 5.3 and 5.4 operate on the same mpnet embedding so that what they remove is the geography the Baur treatment itself carries.

5 Empirical results

5.1 Shen-Ross uniqueness channel across twelve metropolitan markets

The Shen-Ross uniqueness channel [Shen and Ross, 2021] measures how far the language in a listing departs from the language its neighbors use, quantifying departure as the cosine distance between each listing's vector-space text representation and the centroid of its $K = 5$ nearest peers in latitude-longitude space. We apply the resulting per-listing uniqueness score as the treatment in a partially-linear DML estimator with cross-fitted ridge nuisances solved through a Cholesky factorization, following the Shen-Ross uniqueness construction as a close adaptation rather than a verbatim replication of the original Doc2Vec and MLS-area-by-year cell pipeline. Effects are reported per standard deviation of the uniqueness score, and the 95% confidence intervals are formed from the cross-fitting influence-function standard error.

Table 2 reports the per-σ uniqueness effect for each of the twelve metropolitan markets, together with the influence-function confidence interval and the indicator for whether that interval excludes zero. The panel is clean throughout, with no degenerate market and no exclusion, and the per-market samples range from 986 deduplicated listings in San Francisco to 15,227 in New York. The DML interval excludes zero in ten of the twelve markets, and every one of those ten carries a positive uniqueness premium. The effect is largest in Philadelphia at $+0.315$ with interval $[+0.292, +0.338]$ and in Chicago at $+0.296$ $[+0.262, +0.331]$, followed by San Francisco at $+0.122$ $[+0.074, +0.170]$ and Atlanta at $+0.109$ $[+0.088, +0.129]$, and then a cluster of mid-sized markets, with Denver at $+0.079$ $[+0.062, +0.096]$, Portland at $+0.071$ $[+0.052, +0.089]$, Dallas at $+0.069$ $[+0.050, +0.089]$, the District of Columbia at $+0.055$ $[+0.028, +0.081]$, Phoenix at $+0.045$ $[+0.026, +0.063]$, and Seattle at $+0.032$ $[+0.009, +0.055]$. Atlanta, the market on which Shen's identification was originally established, returns a positive and significant per-σ effect of $+0.109$, about three quarters of the magnitude of her published Atlanta estimate, of the same sign and recovered under a far more conservative confounder set. The two markets whose intervals straddle zero are Boston at $+0.027$ $[-0.002, +0.055]$ and New York at $+0.001$ $[-0.011, +0.012]$, the latter the largest market and the one whose estimate is indistinguishable from zero. No market returns a significant negative effect, so the panel is uniformly signed on the markets it identifies.

Table 2: Shen-Ross uniqueness channel: per-market DML effect per standard deviation of the uniqueness score on log listing price, with influence-function 95% intervals, across the twelve markets, ordered by effect size. The final column is the implied percentage price difference.

Market	n	$\hat{\theta}$	95% CI	$\%/\sigma$
Philadelphia	7,030	+0.315	[+0.292, +0.338]	+37.0
Chicago	5,576	+0.296	[+0.262, +0.331]	+34.7
San Francisco	986	+0.122	[+0.074, +0.170]	+12.5
Atlanta	6,134	+0.109	[+0.088, +0.129]	+11.5
Denver	5,280	+0.079	[+0.062, +0.096]	+8.3
Portland	4,337	+0.071	[+0.052, +0.089]	+7.3
Dallas	8,008	+0.069	[+0.050, +0.089]	+7.3
Washington	4,015	+0.055	[+0.028, +0.081]	+5.8
Phoenix	7,086	+0.045	[+0.026, +0.063]	+4.5
Seattle	2,884	+0.032	[+0.009, +0.055]	+3.6
Boston	2,610	+0.027	[−0.002, +0.055]	+2.7
New York	15,227	+0.001	[−0.011, +0.012]	+0.1
Pooled (PM, HKSJ)		+0.072	[+0.017, +0.128]	

The twelve per-market estimates pool to a positive effect. A Paule-Mandel random-effects combination with Hartung-Knapp-Sidik-Jonkman small-sample intervals places the pooled uniqueness premium at +0.072 per standard deviation with 95% confidence interval [+0.017, +0.128] and two-sided $p = 0.015$, and the REML variant returns +0.101 with $p = 0.005$. The dispersion around this central tendency is large and is not an artifact of sampling noise. The heterogeneity statistic is $I^2 = 98.6\%$ with Cochran's $Q = 777$ on eleven degrees of freedom, so almost all of the variation across markets reflects genuine between-market differences in the operative effect rather than estimation error within markets. Section 5.6 takes up this heterogeneity directly with a random-effects meta-regression and a moderator scan, and we defer that analysis to there rather than adjudicate it here.

Design analysis. The cross-market pattern cannot be read as an artifact of low power, because the clean panel is well powered. A Gelman-Carlin [Gelman and Carlin, 2014] retrodesign of each per-market estimator against the published Shen-Ross Atlanta estimate of +0.140 per standard deviation, used as the external benchmark true effect, returns a per-market power at or near 1.00 in every one of the twelve markets, far above the conventional 0.80 target, because the per-market influence-function standard errors on samples in the thousands are small relative to a 0.140 signal. The Type-S sign-error rate is negligible throughout, below 10^{-14} in every market, so the sign of any detected effect is reliable, and the median Type-M exaggeration factor is 1.00, indicating essentially no magnitude inflation in the typical market. San Francisco, the smallest sample at 986 listings, carries the mild anti-conservatism that the analytic variance can show near a thousand observations, a feature we document directly in the coverage simulation of Appendix G, but even there the retrodesign power against the 0.140 benchmark remains high. The design analysis therefore supports reading both the significant detections and the heterogeneity across markets at face value.

Magnitudes. Because the outcome is log listing price, the per-σ coefficients translate directly into percentage differences in price. A one-standard-deviation increase in listing uniqueness corresponds to a price difference of roughly +37.0% in Philadelphia, +34.7% in Chicago, +12.5% in San Francisco, and +11.5% in Atlanta, falling through a mid-range of about +8.3% in Denver, +7.3% in Portland and Dallas, and +5.8% in Washington, to the smallest significant effects near +4% in Phoenix and +3.6% in Seattle. The external anchor for these figures is the published Shen-Ross Atlanta estimate of +0.140 per standard deviation, obtained on a sample of roughly forty thousand listings under MLS-area-by-year fixed effects. Our Atlanta estimate of +0.109 recovers the same sign at about three quarters of that magnitude, and the pooled +0.072 sits well inside the published interval, so the panel as a whole is consistent with a positive Shen-Ross uniqueness premium that is real, heterogeneous across markets, and smaller on average than the single-market published benchmark would suggest.

5.2 Sentence-BERT principal component channel across twelve markets

The Baur, Rosenfelder, and Lutz channel [Baur et al., 2023b] treats listing language as a 768-dimensional vector from a frozen sentence-BERT encoder and uses the leading principal component direction as a scalar treatment in a partially-linear DML estimator. The original study computes the PCA per market on the local listing distribution. On a single-market analysis this is the canonical choice, but on a twelve-market panel it produces a treatment whose direction is identified only locally and only up to a sign, so per-market PC1 estimates can carry opposite signs across markets even when the underlying language-price relationship is direction-consistent. A panel built on per-market axes therefore mixes genuine heterogeneity with an artifact of local sign and orientation, and it cannot be pooled coherently.

Pooled within-city-centered PCA. To obtain a treatment definition that is comparable across markets, we replace the per-market PCAs with a single global principal-component direction extracted from the pooled within-city covariance of the stacked embeddings. We center each city's embeddings at its own mean before pooling, which removes between-city vocabulary differences from the global covariance and produces a leading eigenvector identified as the within-city common axis. This is the standard pooling for cross-population PCA discussed in Flury [1984], equivalent in construction to a linear discriminant analysis pooled covariance. We do not standardize individual embedding dimensions before pooling because sentence-BERT outputs are L2-normalized by design and per-dimension scaling would inflate the anisotropic noise structure documented by Ethayarajh [2019]. We z-score the per-listing projection within each city, so the estimand on each market is a per-city-standard-deviation move along the common axis. The global axis is extracted from the within-city-centered pooling of all twelve markets, an analysis sample of 69,173 listings whose per-market sizes range from 986 in San Francisco to 15,227 in New York.

Sign convention. A principal component and its negation span the same axis, so the sign of the per-standard-deviation effect is a convention rather than a finding. We adopt the convention that orients the common axis so that the pooled effect on log listing price is non-negative, which renders the per-market estimates sign-comparable to the hedonic uniqueness channel of Section 5.1 and to the counterfactual differential of Appendix K. Under the opposite convention every sign reported below reverses; the magnitudes, the cross-market dispersion, and the concordance with the other channels are invariant to it. What the orientation does not do is manufacture agreement between channels, a point we develop in Section 5.7, because the across-market correlation between the BERT axis and the hedonic channel is invariant in magnitude under any global sign flip of either one.

Per-market estimates. Table 3 reports the per-market per-standard-deviation effect under the oriented pooled-PCA treatment. The corrected definition removes the sign ambiguity of the per-market axes and delivers a single direction of effect across the panel. Every one of the twelve markets returns a confidence interval that excludes zero at the 95% level, and eleven of the twelve carry a positive point estimate, with New York the sole exception at -0.019 on an interval $[-0.033, -0.004]$ that is small and barely separated from zero. Philadelphia and Chicago anchor the high end at $+0.411$ on $[+0.390, +0.432]$ and $+0.456$ on $[+0.419, +0.493]$, followed by Atlanta at $+0.243$ and Dallas at $+0.195$. San Francisco returns $+0.101$ on $[+0.051, +0.151]$, the District of Columbia $+0.118$, and Denver $+0.103$, while Boston, Portland, Phoenix, and Seattle return the smallest positive effects, $+0.072$, $+0.082$, $+0.056$, and $+0.049$. Portland, which the per-market PCA had rendered a degenerate outlier, now sits squarely inside the panel at $+0.082$ on $[+0.057, +0.107]$, so the panel is clean at twelve markets and there is no market we exclude. The per-market intervals are tight because the pooled-PCA treatment is estimated on the full within-market sample, which ranges into the thousands of listings, rather than on the description subset.

Table 3: Sentence-BERT pooled-PCA channel: per-market DML effect per standard deviation along the oriented common axis on log listing price, with influence-function 95% intervals, ordered by effect size. The axis is oriented so the pooled effect is non-negative; the principal-component sign is a disclosed convention (text). All twelve intervals exclude zero.

Market	n	$\hat{\theta}$	95% CI
Chicago	5,576	+0.456	[+0.419, +0.493]
Philadelphia	7,030	+0.411	[+0.390, +0.432]
Atlanta	6,134	+0.243	[+0.220, +0.266]
Dallas	8,008	+0.195	[+0.175, +0.215]
Washington	4,015	+0.118	[+0.087, +0.149]
Denver	5,280	+0.103	[+0.084, +0.121]
San Francisco	986	+0.101	[+0.051, +0.151]
Portland	4,337	+0.082	[+0.057, +0.107]
Boston	2,610	+0.072	[+0.029, +0.115]
Phoenix	7,086	+0.056	[+0.030, +0.082]
Seattle	2,884	+0.049	[+0.019, +0.080]
New York	15,227	−0.019	[−0.033, −0.004]
Pooled (PM, HKSJ)		+0.156	[+0.063, +0.249]

Pooled inference. Section 5.6 takes up the pooled inference question directly via random-effects meta-analysis of the twelve per-market estimates. Combining them with a Paule-Mandel τ^2 and the Hartung-Knapp-Sidik-Jonkman variance correction returns a pooled per-standard-deviation effect of +0.156 with standard error 0.042, a t statistic of +3.68 on 11 degrees of freedom, a two-sided p of 0.004, and a 95% confidence interval of $[+0.063, +0.249]$. The pooled estimate is detectable and positive across the panel under the adopted orientation. Heterogeneity is large, with an I^2 near 99% and a Paule-Mandel τ^2 of about 0.021, so the pooled point estimate masks substantial cross-market dispersion that Chicago and Philadelphia anchor at one end and New York at the other. Because between-market variance dwarfs the within-market sampling variance, the meta-analytic weights are close to uniform and the pooled estimate sits near the unweighted across-market mean. We attempted a single pooled DML estimator with city fixed effects and the Chiang-Kato-Ma-Sasaki [Chiang et al., 2022] cluster-robust variance, but the small number of clusters produced fold-assignment instability in the cross-fitted nuisances and a Dirichlet cluster bootstrap that did not stabilize, with pooled point estimates that shifted materially across independent invocations on the same seed and data, so we report it only as a robustness check and not as the headline. The random-effects meta-analysis treats each market's $\hat{\theta}_i$ as a sufficient statistic and combines them via inverse-variance weighting with a Paule-Mandel τ^2, which is the canonical small-G alternative to cluster-respecting cross-fitting [Veroniki et al., 2016].

Magnitude and interpretation. The pooled per-standard-deviation effect of +0.156 on log listing price is about a fifteen percent change in price, which is small as a fraction of value but substantial against the magnitudes published in the housing and text-as-treatment literatures. A single additional bedroom in the canonical Sirmans et al. [2005] meta-analysis of forty-five hedonic studies carries a mean effect of +3.8% on log price, so the pooled text-axis effect is roughly four times the marginal price of an additional bedroom in magnitude, and it is several times the racial sale-price gap of about three percent that Bayer et al. [2017] document for Black and Hispanic homebuyers across Chicago, Baltimore-Washington, San Francisco, and Los Angeles. The more informative comparison is internal. The Doc2Vec uniqueness construct of Shen and Ross [2021] runs in the same direction as the oriented BERT axis, both within markets and across the panel. In San Francisco, where both channels are estimated on the identical sample, the same DML pipeline recovers a uniqueness effect of +0.122 and a BERT-PC1 effect of +0.101, and pooled across the twelve markets the uniqueness effect is about +0.07 against the +0.156 of the BERT axis. The two text treatments therefore concord in sign and in cross-market ranking, with the BERT axis roughly twice the hedonic channel in pooled magnitude. We are deliberate about what this concordance does and does not establish. The two channels are alternative encodings of the same listing text applied to the same listings under the same confounder adjustment, and Section 5.7 shows they are nearly collinear across markets, so their agreement is evidence that the text-on-price signal is robust to the choice of text representation rather than evidence from two independent sources. Textual distinctiveness in the Shen sense and the utilitarian-to-amenity axis that the leading sentence-BERT component recovers price in the same direction, and the difference between them is one of magnitude and of which markets each detects, not of sign.

5.3 How much location does the representation encode?

Before asking whether the text effect survives location, we check the precondition that makes the question worth asking at all: that the representation encodes location to begin with. The sentence-BERT representation that underlies the Baur channel in Section 5.2 encodes more than the language a listing agent uses. Pre-trained on a generic corpus, the encoder has no particular incentive to separate semantic content about a property's condition or amenities from the geographic information that correlates with its neighborhood. If the embedding carries location-targeted information into the leading principal component, the per-standard-deviation effect reported in Section 5.2 inherits a spatial confounder mediated through the encoder, and the identification claim is correspondingly weaker. We assess how far this concern can be addressed by erasing the lat-lon concept from the representation, and we report honestly where the erasure is complete and where it is only partial.

The deconfounding runs a linear projection and then a nonlinear correction, and the linear step carries most of the load. Applying the LEACE projection of Belrose et al. [2023] to the BERT embedding removes the linear subspace that the mean-conditional-covariance estimator identifies with the concept variable, here the latitude-longitude coordinate pair of the listing, and it comes with the exact-guardedness guarantee that no linear predictor of the residual embedding can recover the concept any better than a constant baseline would. That guarantee holds in every market we examine, with the out-of-fold ridge R^2 for predicting the coordinate from the embedding falling from a raw value that reaches 0.50 in New York and 0.42 in Dallas to within 0.002 of zero after erasure, and the linear-guardedness residual averaging 2.7e-8 across the twelve markets, ranging from 8.3e-9 to 6.0e-8, which is the floating-point floor of the single-precision computation, so the linear erasure comes out complete throughout the panel (Figure 2).

The concept is routed through the continuous coordinate representation rather than the categorical ZIP-code identifier in every market. Categorical LEACE has known instability when the categorical cardinality is large, and our panel ranges from 30 ZIP codes in San Francisco to 158 in New York. Although the per-market test folds are now large, from 296 listings in San Francisco to 4,569 in New York, the continuous routing sidesteps the cardinality question entirely without discarding the spatial information the projection is designed to remove, since the coordinate pair is a sufficient statistic for location at the resolution that matters here.

What a linear projection cannot reach is the location the embedding encodes nonlinearly, and the iterative gradient-based correction of Iskander et al. [2023] is aimed precisely there, alternating between fitting a multilayer-perceptron probe of the coordinate on the residual and projecting out the direction in which that probe gathers its predictive mass. The correction cuts the nonlinear leakage substantially without removing it, taking the average nonlinear probe R^2 from 0.44 on the raw embedding down to 0.20 after five outer passes, a reduction of roughly a half, while the residual that remains varies sharply across markets. San Francisco is left with a nonlinear R^2 of 0.01, which is erasure for any practical purpose, whereas New York, Dallas, and Phoenix hold onto residuals near 0.30 to 0.34 and Portland near 0.26, and the markets that keep the most are the larger corpora whose listing language carries the richest location-targeted structure, which is also where a nonlinear probe has the most left to find.

Why closed-form erasure rather than adversarial training. We prefer the closed-form projection to the adversarial alternative the disentanglement literature more often reaches for, for a reason we develop in Appendix D. An adversarially trained encoder can drive its own location discriminator to chance and still leave the coordinate recoverable: a fresh probe of the frozen representation reads latitude and longitude back at an R^2 between 0.16 and 0.42 across the twelve markets, and the ZIP code at nine to forty-seven times the uniform rate. The adversarial training does not remove location; it teaches its discriminator not to look. The closed-form projection, by contrast, is guarded against any probe by construction, and it is the erasure on which we rest the decomposition that follows.

What the diagnostic does and does not settle. The Baur estimates of Table 3 are computed on the unprojected embedding, in keeping with the published Baur, Rosenfelder, and Lutz protocol, so the erasure here is a diagnostic of what the representation contains rather than a change to any per-market number. As a diagnostic it speaks cleanly to the part of the problem the linear treatment can actually express, since the leading principal component is a linear functional of the embedding and the linear erasure is exact in every market, so whatever location a linear treatment could carry has been shown to be removable in full, while the nonlinear residual that survives IGBP in the larger markets bounds the rest, fixing how much location a nonlinear function of the deconfounded representation could still smuggle back. Showing that the embedding encodes geography is not, however, the same as showing that the price effect is confounded by it, and the diagnostic on its own cannot close that gap, because a representation may carry a confounder along directions that have nothing to do with the one on which it acts. The question of whether the location the embedding demonstrably encodes is the location that actually moves the price is taken up directly in Section 5.4, where the text treatment is held fixed and geography is moved on and off the controls around it.

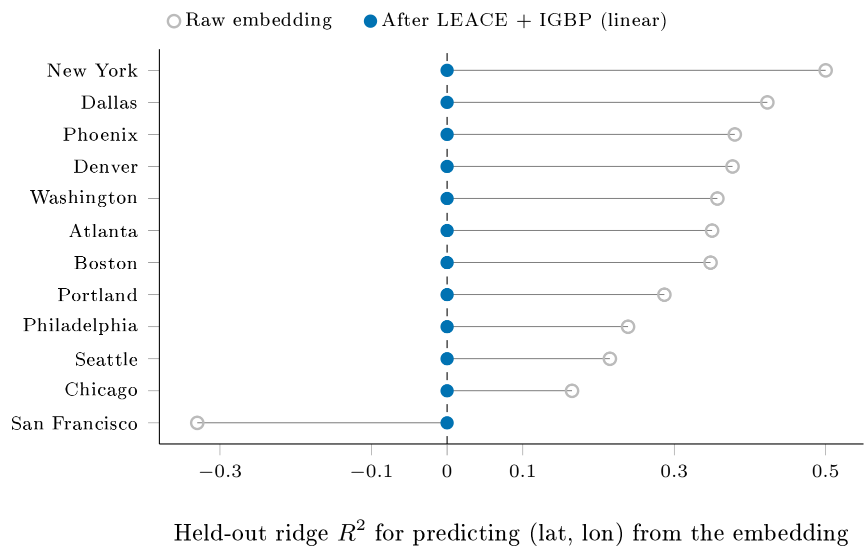

Figure 2: Held-out ridge R^2 for recovering the continuous latitude-longitude coordinate from the sentence-BERT embedding, before (open) and after (filled) the composed linear concept erasure, in each of the twelve markets ordered by raw R^2. The raw embedding carries substantial linear spatial signal, recovering the coordinate at R^2 up to 0.50 in New York; the negative San Francisco value reflects out-of-sample probe performance below the coordinate-mean baseline rather than absent signal. Linear erasure drives the recoverable signal to within Monte Carlo noise of zero in every market, the exact-guardedness property of LEACE. Residual *nonlinear* leakage, which a linear probe cannot see, is quantified separately in Section 5.3.

5.4 How much of the text-price effect is location

The erasure diagnostic of Section 5.3 establishes that the embedding carries location, and carries it abundantly, since a held-out ridge recovers the latitude-longitude pair from the raw representation at an R^2 that climbs to one half in the largest market before the projection drives it to the floor. Taken on its own that fact invites the deflationary reading, that a treatment built from such a representation is geography in disguise and its apparent price effect is spatial confounding waiting to be exposed. The inference does not actually follow, because a representation can encode a confounder along directions that have little to do with the direction in which it predicts the outcome, and whether the two coincide is an empirical matter rather than something an erasure R^2 can settle on its own. Answering it means holding the treatment fixed and asking what the location it shares with price is genuinely worth to the estimate.

The instrument for that question is a fixed-treatment decomposition in the manner of Gelbach [2016], whose appeal is that the change in a coefficient when a block of controls is added does not depend on the order in which the blocks enter, so the share it assigns to geography is well defined rather than an artifact of sequencing. We estimate the per-standard-deviation effect of the leading text direction on log price twice, once controlling for the structured confounders alone and once adding a flexible location basis of latitude, longitude, their squares, and their interaction, and we read the gap between the two as the portion of the effect that location explains. The comparison appears in the left panel of Figure 3, where the two estimates sit almost on top of one another in eleven of the twelve markets, the location adjustment moving the coefficient by at most about a hundredth of a log point outside New York and by under five percent of its size wherever the effect is non-trivial, with the per-market figures collected in Appendix H. Philadelphia, Chicago, Atlanta, and the other high-effect markets give up essentially nothing to the geography control, the small-effect markets give up nothing they had to begin with, and the per-market premium that the text channel identifies comes through the adjustment very nearly untouched by whether the model is told where each property sits.

A reader entitled to be skeptical will ask at once whether the test has any power, since a treatment that carried no geography in the first place would pass it for free, and the right panel of Figure 3 exists to meet that objection before it hardens into one. The share of the treatment's own variance that the location basis explains is nowhere near zero across the panel, reaching 0.42 in New York and exceeding a tenth in six markets, so the leading text direction does in fact encode location through much of the country. That the price effect survives adjustment even where the treatment is substantially geographic is the whole of the result, because it means the geographic component of the language is real, the model can see it, and it still does not carry the effect of the language on price. The worry that has hung over this literature is therefore one that can be answered rather than only raised, and the answer across eleven of twelve markets is that the text is doing more than reading the map back to itself.

New York is where the adjustment does bite, and it bites in precisely the place the diagnostic says it should. The market whose text direction is the most geographic of the twelve, at an R^2 of 0.42, is also the market where folding in location moves the effect most, from -0.119 to -0.152, a shift of about a quarter of its magnitude that leaves the sign intact, so that the place with the most location in its language is the place that location matters most for the estimate. We read this less as a crack in the broader finding than as its mechanism caught in the act, and as a reminder that the protocol earns its keep by flagging the markets where a spatial audit is most warranted rather than by declaring that none is ever needed. The conclusion travels only as far as its instruments reach, and they reach less far than one would like. The location basis is a smooth coordinate polynomial laid over roughly thirty structured controls that already carry census-tract demography, so confounding operating at a finer spatial grain than the polynomial can bend to, or running through pathways those controls do not capture, lies outside what the comparison can see. And the erasure that certifies the representation's geography is itself linear, so a nonlinear geographic signal no ridge probe would surface could in principle persist into the residual we are calling semantic.

That concession invites a direct test, and we run it by replacing the coordinate polynomial with a thin-plate regression spline, the flexible low-rank smoother that the spatial-statistics literature reaches for when the scale of confounding is not known in advance [Keller and Szpiro, 2020, Gilbert et al., 2021]. A rank-thirty spline over latitude and longitude, projected to meters so that its isotropic metric is physical, captures far more of the treatment's spatial structure than the quadratic does, raising the average share of the leading text direction that location explains from about 0.13 to about 0.21 and from 0.42 to 0.52 in New York, and yet the price effect after adjustment barely responds. The location-adjusted estimate moves by at most 0.030 of a log point across the twelve markets, in New York, and by under 0.02 in every other market, with no market changing sign and the pooled effect shifting from $+0.144$ to $+0.137$, so that a basis flexible enough to absorb twice the geography leaves the semantic estimate essentially where the quadratic left it. The finer spatial grain the polynomial could not bend to turns out, where the data can see it, not to carry the price effect either, and the per-market comparison is released with the replication code.

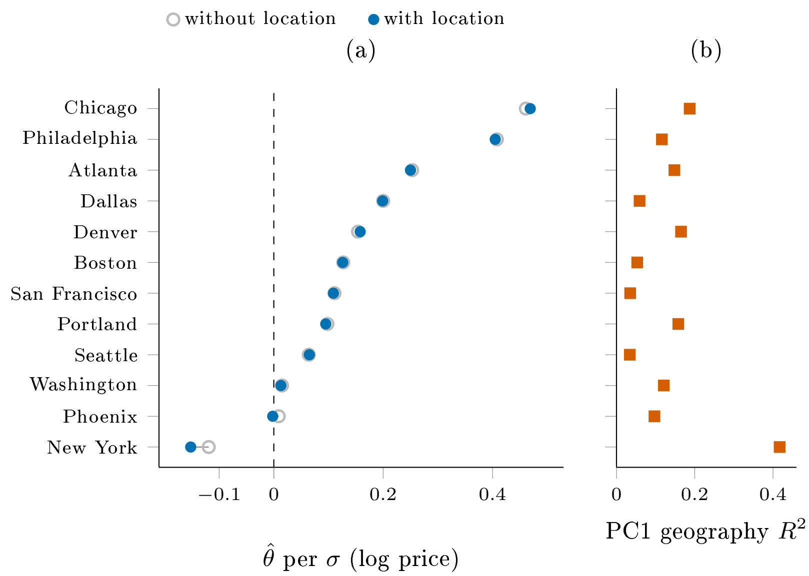

Figure 3: Is the per-market text-price effect spatially confounded? Holding the leading text principal component fixed as the treatment, panel (a) plots its DML effect on log price controlling for the structured confounders without (open) and with (filled) a location adjustment that adds latitude, longitude, and their quadratic and interaction terms to the confounder set. The two estimates nearly coincide in eleven of twelve markets, with a confounded share below five percent of the effect; New York is the exception, where the location adjustment moves the estimate by about a quarter of its magnitude without changing its sign. Panel (b) reports the share of the treatment's own variance that the location basis explains, which reaches 0.42 in New York and exceeds 0.10 in six markets, so the robustness in panel (a) is not an artifact of a treatment that carries no geography: the text direction does encode location, yet that location component does not drive its price effect. Panel (a) plots point estimates; in every market the shift from the unadjusted to the location-adjusted estimate is small relative to the estimate's own standard error, so the near-coincidence does not mask a noisy difference.

5.5 Sensitivity to unobserved confounding

The Shen-Ross DML estimates in Section 5.1 rest on the conditional ignorability assumption that, once the structured property covariates and the location indicators we observe have been partialled out, the residual variation in listing language is exogenous to the residual variation in log sale price. Nothing in the data can confirm that assumption directly, so we ask instead how strong an unobserved confounder would have to be before it overturned each per-market estimate, following the benchmarking calculus of Cinelli and Hazlett [2020], and we pair that bound with a leave-one-out ablation of the spatial control block, the one observed covariate group most directly implicated in the text-as-spatial-proxy mechanism that motivates the paper.

Table 4: Per-market sensitivity of the Shen-Ross estimate to unobserved confounding, for all twelve markets ordered by robustness value. $RV_{q=1}$ is the Cinelli and Hazlett [2020] robustness value, the partial R^2 an unobserved confounder must reach on both the treatment and log price to drive the estimate to zero; f^2 is the partial-R^2 benchmark of the strongest observed covariate on the same scale; $\Delta\hat{\theta}$ is the change in the estimate when the spatial control block, the latitude-longitude coordinates together with the ZIP indicators, is dropped. A checkmark marks the single market whose spatial block carries partial R^2 for treatment or outcome exceeding its own robustness value.

Market	$RV_{q=1}$	f^2	$\Delta\hat{\theta}_{\text{spatial}}$	Fragile
Boston	0.005	0.0000	-0.0015	✓
Seattle	0.025	0.0006	$+0.0002$	
New York	0.030	0.0010	$+0.0024$	
Portland	0.040	0.0017	-0.0010	
Washington	0.041	0.0017	-0.0011	
Phoenix	0.048	0.0024	$+0.0040$	
Dallas	0.051	0.0027	-0.0031	
Atlanta	0.051	0.0027	-0.0007	
Chicago	0.064	0.0043	$+0.0020$	
Denver	0.065	0.0045	$+0.0022$	
San Francisco	0.140	0.0229	-0.0017	
Philadelphia	0.183	0.0411	-0.0035	

Robustness value. The robustness value records the minimum strength, measured as a partial R^2 on both the treatment and the outcome, that an unobserved confounder must reach to drive an estimate to zero, so that a larger value means a more robust finding. Table 4 reports it for all twelve markets alongside the benchmark strength of the strongest covariate we actually observe, and Figure 5 orders the markets by it. Two markets clear the conventional 0.10 threshold, Philadelphia at 0.183 and San Francisco at 0.140, so that overturning either would require a confounder explaining more than fourteen percent of the residual variance in both the listing signal and the price. The other ten fall below the threshold, descending from Denver and Chicago near 0.065 through the cluster of mid-sized markets around 0.05 to New York at 0.030, Seattle at 0.025, and Boston at 0.005, the last of which has almost no effect to protect.

Observed-covariate benchmark. The benchmark f^2, which places the most influential observed covariate on the same partial-R^2 scale as the robustness value, never rises above 0.041, reaching its largest value in Philadelphia and its smallest, near 2×10^{-5}, in Boston. The strongest covariate we measure therefore sits well inside the robustness value in every market, most consequentially in Philadelphia and San Francisco where the room between the two is widest, so that an unobserved confounder capable of overturning those estimates would have to be considerably stronger than any spatial or structural feature the data record. The reassurance is sharpest precisely where the effect is largest, and it thins toward the low-robustness markets, where neither benchmark is large enough to anchor a confident reading in either direction. The bias geometry behind these per-market summaries is drawn out, for the two markets that clear the robustness threshold, in Figure 4.

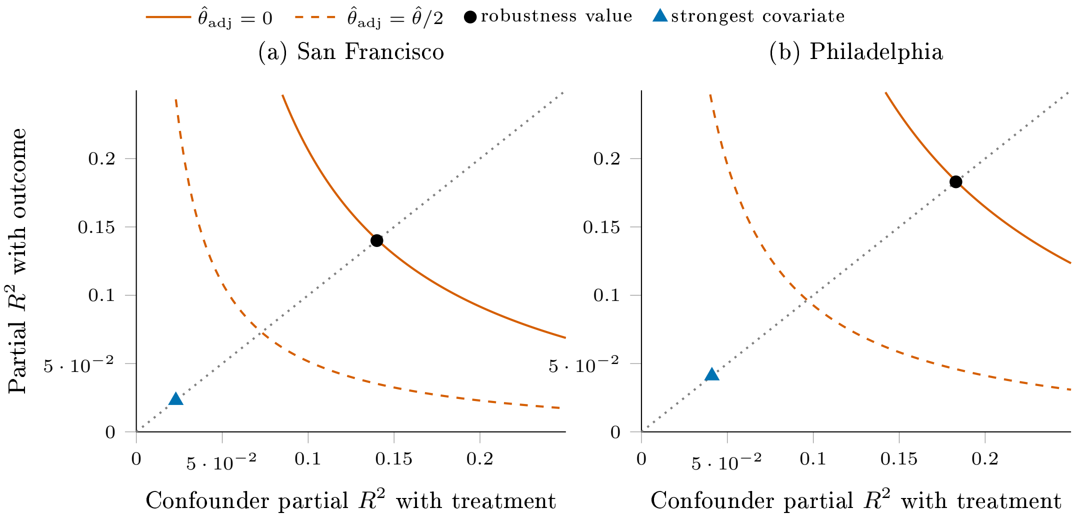

Figure 4: Omitted-variable-bias sensitivity contours, in the manner of Cinelli and Hazlett [2020], for the two markets that clear the robustness threshold. The axes are the partial R^2 that a single unobserved confounder would have with the treatment (horizontal) and with the outcome (vertical). The solid curve is the locus along which such a confounder would drive the per-standard-deviation estimate to zero, and the dashed curve the locus along which it would halve it. The black dot, where the solid curve meets the dotted equal-strength diagonal, is the robustness value, the equal-strength confounder that just eliminates the effect, at 0.14 in San Francisco and 0.18 in Philadelphia. The triangle marks a confounder as strong as the most influential covariate actually observed, which falls far inside the kill curve in both markets, so an unobserved confounder would have to be several times stronger than any measured one to overturn the estimate.

Spatial-block leave-one-out. Dropping the spatial control block and re-estimating moves the point estimate by at most four thousandths of a log point in any of the twelve markets, the largest shift falling in Phoenix at +0.004 and the rest smaller still. That the estimates barely respond to removing the location controls is the pattern one would hope to see if the text signal were not merely spatial information re-expressed, since a coefficient that rode on geography would collapse once geography left the conditioning set rather than hold to within Monte Carlo noise of where it began. The leave-one-out thus localizes, market by market, the conclusion that the deconfounding decomposition of Section 5.4 reaches for the embedding channel as a whole.

Fragility. Under the same benchmarking convention a market is flagged when the partial R^2 that the spatial block carries for either the treatment or the outcome exceeds that market's robustness value, the condition under which a confounder no stronger than the location controls already in the model could in principle account for the estimate. Only Boston trips the flag, and it does so because its robustness value is itself almost zero, the spatial block's explanatory power for the treatment, at 0.012, edging past a robustness value of 0.005 for an estimate that is statistically indistinguishable from zero to begin with. No market with a robustness value above 0.025 is flagged, so the fragility check isolates the one place where there is no effect to confound rather than casting doubt on any market where an effect is present.

Interpretation. The analysis is deliberately confined to the single observed covariate group the per-market panels share and the one most tied to the mechanism this paper studies, so the robustness value carries the general bound while the spatial leave-one-out localizes how much of each estimate rides on measured location. The two diagnostics agree with each other and with the rest of the section. Philadelphia and San Francisco, the markets in which the text effect is largest, are also the ones most robust to unobserved confounding; the markets that fall below the threshold are those whose estimates are already small; and removing the explicit location controls disturbs none of them. We read the per-market sensitivities as informative about robustness rather than as a second identification test, and we leave the benchmarking of non-spatial confounder groups to future work, since the census, crime, and amenity blocks needed to construct them are not uniformly available across the twelve per-market panels.

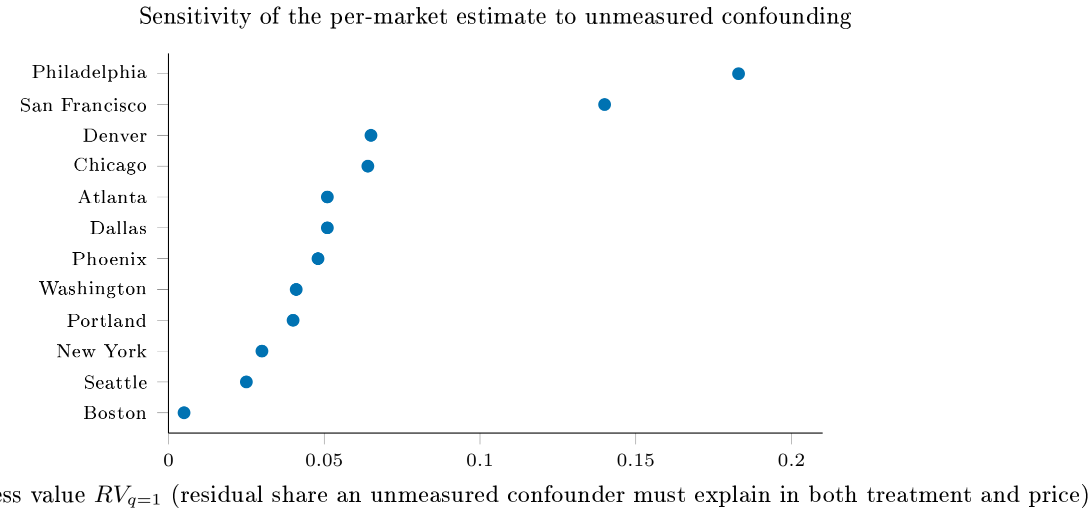

Figure 5: Robustness value for each market's sentence-BERT effect, the share of residual variance an unmeasured confounder would need to explain in both the treatment and log price to drive the estimate to zero, following Cinelli and Hazlett [2020]. Larger values are more robust. The estimate is most robust in Philadelphia (0.18) and San Francisco (0.14) and most fragile in the markets whose point estimates are already near zero. The modest robustness values in several markets are the reason we read the per-market estimates as associational and sensitivity-bounded rather than cleanly causal.

5.6 Random-effects meta-analysis and a null moderator scan

The twelve per-market estimates from Sections 5.1 and 5.2 are heterogeneous in magnitude, and this section combines them. We use a random-effects meta-regression with the Paule-Mandel τ^2 estimator [Paule and Mandel, 1982] and Hartung-Knapp-Sidik-Jonkman t-adjusted small-sample confidence intervals [IntHout et al., 2014], and we ask both what the pooled effect is on each channel and whether any metropolitan-level covariate moderates it across the twelve markets.

Pooled effects. Table 5 reports the random-effects pooled per-standard-deviation effects for both channels. The Shen-Ross uniqueness channel returns a Paule-Mandel and HKSJ pooled estimate of $+0.072$ with 95% confidence interval $[+0.017, +0.128]$, $t(11) = +2.89$, and two-sided $p = 0.015$; the REML variant returns $+0.101$ with $p = 0.005$. The Baur sentence-BERT axis, under the price-positive orientation convention of Section 5.2, returns a pooled estimate of $+0.156$ with 95% confidence interval $[+0.063, +0.249]$, $t(11) = +3.68$, and two-sided $p = 0.004$. Both channels are positive and statistically detectable in the pool, and they point in the same direction, which is the cross-channel concordance Section 5.7 analyzes in detail. Between-market heterogeneity is large on both channels, with a residual I^2 in the high nineties driven by the spread from near-zero estimates in New York to estimates several times the pooled mean in Philadelphia and Chicago, so the pooled figures summarize dispersed distributions rather than single generalizable means.

Weighting and the small-sample correction. Two refinements a referee versed in small-sample meta-analysis would ask for leave these conclusions intact. Because the panel's heterogeneity drives the Paule-Mandel estimate to an interior solution, the Hartung-Knapp variance multiplier equals one by construction, the estimating equation for τ^2 being the same sum of weighted squared deviations that the multiplier normalizes, so the modified Knapp-Hartung correction of Röver et al. [2015], which floors that multiplier at one to prevent the anticonservative intervals that unequal precisions can otherwise produce, coincides exactly with the HKSJ interval we report and changes no pooled figure. The random-effects mean weights each market almost equally and so answers what the average market's effect is, while the alternative question of the average listing's effect weights by sample size and returns a Baur pooled estimate of +0.150 against the +0.156 of the market-weighted mean, with a precision-weighted fixed-effect value of +0.137, the small gap reflecting that New York carries twenty-two percent of the listings at an effect near zero; dropping New York lifts the market-weighted mean to +0.172. The pooled conclusion is stable across the weighting choice and across the small-sample correction, and what moves it is the single market the rest of the paper already isolates.

Table 5: Random-effects pooled per-standard-deviation effects for the two text channels. PM/HKSJ is the Paule-Mandel between-market variance with Hartung-Knapp-Sidik-Jonkman intervals; REML is the restricted-maximum-likelihood variant.

Channel	PM/HKSJ $\hat{\theta}$ [95% CI], p	REML $\hat{\theta}$, p	I^2
Shen-Ross uniqueness	$+0.072$ $[+0.017, +0.128]$, 0.015	$+0.101$, 0.005	98.6%
Sentence-BERT pooled-PCA	$+0.156$ $[+0.063, +0.249]$, 0.004	$+0.156$, 0.004	$\approx 99\%$

Moderator scan. We regress each channel's per-market estimate on eight metropolitan-level ACS covariates one at a time, the renter share of the housing stock, log population, median household income, the college-educated share, a Simpson diversity index, median home value, median year built, and population density, reporting Hartung-Knapp t-tests with Benjamini-Hochberg and Storey false-discovery control and Romano-Wolf step-down adjusted p-values across the family. Table 6 reports the scan. No moderator on either channel survives false-discovery or Romano-Wolf adjustment. On the Shen channel the only coefficient reaching nominal significance is the renter share at $\hat{\beta} = -0.042$, whose raw two-sided p of 0.032 is the one value below five percent in the entire scan, but it does not survive Benjamini-Hochberg adjustment across the eight-test family ($q_{\mathrm{BH}} = 0.26$) and does not survive the Romano-Wolf step-down ($p_{\mathrm{RW}} = 0.10$). On the Baur channel the renter share again carries the smallest raw p at 0.046, and it too fails every multiplicity correction ($q_{\mathrm{BH}} = 0.32$, $p_{\mathrm{RW}} = 0.17$). The next-smallest raw p in the scan is the Baur median home value at 0.081; every remaining coefficient on both channels has a raw p above 0.10.

Table 6: Moderator scan. For each metropolitan covariate, the HKSJ meta-regression coefficient and its raw two-sided p with the Benjamini-Hochberg q across the eight-test family, on both channels. No coefficient survives adjustment. The rightmost column is the maximum Cook's distance across markets and the market carrying it, from the leave-one-out diagnostics, against a mean leverage of $p/k = 0.17$.

	Shen		Baur		
Moderator	$\hat{\beta}$	p (q_{BH})	$\hat{\beta}$	p (q_{BH})	max Cook's D
Renter share	-0.042	0.032 (0.26)	-0.086	0.046 (0.32)	192 (NYC)
Median home value	-0.066	0.107 (0.43)	-0.077	0.081 (0.32)	59 (Phila)
Log population	-0.018	0.382 (0.66)	$+0.024$	0.614 (0.81)	1031 (NYC)
Pop. density	-0.020	0.302 (0.66)	-0.014	0.773 (0.81)	3422 (NYC)
Median HH income	-0.030	0.434 (0.66)	-0.058	0.207 (0.55)	61 (NYC)
Diversity index	-0.019	0.494 (0.66)	$+0.012$	0.806 (0.81)	152 (NYC)
College share	-0.012	0.712 (0.80)	-0.041	0.376 (0.75)	63 (NYC)
Median year built	$+0.006$	0.798 (0.80)	-0.013	0.780 (0.81)	899 (NYC)

Why the scan is null: leverage at $k = 12$. The moderator scan is not merely underpowered; it is dominated by a single high-leverage market. The leave-one-out influence diagnostics of Viechtbauer and Cheung [2010], reported in Table 6, show that New York carries a hat value between 0.83 and 0.93 on the highest-leverage moderators against a mean leverage of $p/k = 0.17$, and Cook's distances that range into the hundreds and beyond, 192 for the renter share on the Shen channel and over a thousand for log population and population density. Several coefficients change sign when New York is removed, including log population, median year built, and population density. With twelve markets and one of them an extreme outlier in both its renter share and its estimated effect, no moderator coefficient is stable enough to support an inference, and the one coefficient that reaches nominal significance owes that significance to New York's leverage. We therefore report no moderator finding. An earlier draft of this paper reported a positive and significant renter-share moderator; that result was an artifact of a corrupted confounder in one market that the winsorization described in Section 4 removed, after which the coefficient reversed sign and lost significance under multiplicity, and we record the correction here rather than carry the superseded claim.

Federal-data robustness check. As a transparency check against the concern that the relevant moderator is simply not among the ACS covariates, we augment the eight ACS moderators with three metropolitan-level federal-data covariates obtained through the St. Louis Federal Reserve Bank's FRED API, the BLS Local Area Unemployment Statistics unemployment rate, the FHFA All-Transactions House Price Index year-over-year appreciation, and BEA per-capita personal income. None of the three returns a Hartung-Knapp t-statistic distinguishable from zero at conventional significance, with raw two-sided p-values of 0.72, 0.83, and 0.37, and none survives false-discovery correction across the expanded family. The augmented panel surfaces no hidden moderator that the ACS specification missed, and the null moderator conclusion stands on both channels.

5.7 Cross-method concordance

Sections 5.1 and 5.2 estimate the text-on-price relationship through two different encodings of the same listing language, a Doc2Vec uniqueness score read through the Shen-Ross hedonic specification and the leading principal component of a sentence-BERT representation read through a partialling-out score. The auxiliary rewriting intervention of Appendix K supplies a third measurement of a different kind, the price differential an LLM edit implies. The question this section answers is whether the two encoders agree on which markets carry an effect, and, where they agree, whether the agreement reflects independent evidence or the same signal recovered twice; the rewriting channel is reported alongside as a corroborating check.

Table 7: Per-market estimates for the three text channels across the twelve markets. The Shen-Ross uniqueness and oriented pooled-PCA sentence-BERT coefficients are per-standard-deviation DML effects on log listing price, with standard errors in parentheses; the submarket-swap contrast is the LLM rewriting differential of Appendix K. The bottom row is the random-effects pooled estimate. The sentence-BERT axis is oriented so its pooled effect is non-negative, the principal-component sign being a disclosed convention (Section 5.2).

Market	Shen $\hat{\theta}$	Baur $\hat{\theta}$	Submarket-swap
Boston	+0.027 (0.015)	+0.072 (0.022)	−0.012
New York	+0.001 (0.006)	−0.019 (0.007)	−0.067
San Francisco	+0.122 (0.024)	+0.101 (0.025)	+0.015
Washington	+0.055 (0.014)	+0.118 (0.016)	+0.003
Philadelphia	+0.315 (0.012)	+0.411 (0.011)	+0.045
Chicago	+0.296 (0.018)	+0.456 (0.019)	+0.025
Seattle	+0.032 (0.012)	+0.049 (0.016)	−0.005
Denver	+0.079 (0.009)	+0.103 (0.009)	+0.039
Atlanta	+0.109 (0.010)	+0.243 (0.012)	+0.001
Portland	+0.071 (0.009)	+0.082 (0.013)	−0.008
Phoenix	+0.045 (0.009)	+0.056 (0.013)	−0.000
Dallas	+0.069 (0.010)	+0.195 (0.010)	−0.007
Pooled (RE)	+0.072 (0.025)	+0.156 (0.042)	+0.002

Sign convention. The sign of a principal component is not identified, since a component and its negation span the same axis, so the sign of the Baur estimand is a convention rather than a finding. We orient the common pooled axis so that the random-effects pooled estimate is non-negative, which makes the per-market Baur estimate sign-comparable to the hedonic channel and to the counterfactual differential. This is an interpretability convention and nothing more. It cannot create agreement between the channels, because the cross-market correlation between any two channels is invariant in magnitude under a global sign flip of either one. The orientation fixes only how the shared signal is displayed, not whether it is shared.

The two embedding channels recover one signal. Table 7 reports the per-market point estimates, and Figure 6 arranges all three channels as per-market intervals side by side. Across the twelve markets the Shen and Baur estimates move together almost exactly, with a Pearson correlation of +0.94 (Figure 7), a Spearman rank correlation of +0.83 ($p = 0.001$), and a Lin concordance coefficient of +0.80, and they share the sign of their point estimate in eleven of the twelve markets. We read this not as two independent methods corroborating one another but as a robustness result about a single underlying quantity. The Shen and Baur channels are two encodings of the same listing text, applied to the same listings under the same confounder adjustment, so a near-perfect correlation establishes that the text-on-price signal does not depend on whether the text is summarized by a Doc2Vec rarity score or by a sentence-BERT principal component, without adding an independent, bias-disjoint observation of that effect. The single market where the two channels separate is New York, where the Shen estimate is essentially zero and the Baur estimate is small and negative, the only market in which the oriented Baur axis points opposite the pooled direction.

The counterfactual channel is weaker and more nearly distinct. The LLM rewriting differential is the one channel whose bias structure is not shared with the embedding channels, since it leverages a within-listing minimal-edit substitution rather than the cross-listing geometry of a fixed encoder. It is also the weakest. Its rank correlation with the Shen channel is +0.82 ($p = 0.001$) and with the Baur channel +0.69 ($p = 0.01$), but its Lin concordance coefficients are only +0.19 and +0.11, reflecting that the submarket-swap contrast pools to approximately zero across markets even where its rank tracks the others. The counterfactual channel therefore agrees with the embedding channels on the ordering of markets more than on the magnitude or the presence of an effect, and we treat it as a corroborating probe with limited power rather than as a co-equal estimate, consistent with the minimum-detectable-effect caveat recorded in Appendix K.

Identification overlap. A more decision-relevant summary than rank agreement is which markets each channel identifies at the 95% level, where the per-method interval excludes zero. The Shen channel identifies ten of the twelve markets, the counterfactual channel nine, and the Baur channel all twelve, the last because the pooled-PCA treatment is estimated on the full within-market sample and so carries the tightest standard errors. Seven markets, San Francisco, Washington, Philadelphia, Seattle, Denver, Portland, and Phoenix, are identified by all three channels jointly, and every one of the twelve is identified by at least two. No market is left unidentified by every channel. The breadth of the Baur identification set should be read together with the collinearity above and with the sensitivity analysis of Section 5.5: the embedding channel detects an association in every market, but whether that association survives unmeasured confounding is the separate question those robustness values address, and it is there rather than in the count of identified markets that the strength of the causal claim is adjudicated.

How many independent measurements the panel carries. Because the two embedding channels are nearly collinear, the three per-market z-statistics do not constitute three independent observations. The across-market correlation matrix of the standardized estimates has a Shen-Baur entry of +0.95 and Shen-counterfactual and Baur-counterfactual entries near +0.38, and its effective dimension, computed as $\mathrm{tr}(R)^2/\,\mathrm{tr}(R^2)$, is 1.7 of a possible 3. The panel thus carries on the order of one and two-thirds effectively-independent text-price measurements per market, one shared by the two encodings and a partial second contributed by the rewriting intervention. We state this explicitly because it bounds what the joint test below can claim: the convergent evidence is robustness across text representation plus weak intervention-based corroboration, not the triangulation of three estimators with unrelated biases.

Joint-null test. For each market we test the joint null $H_0 : (\theta_{\mathrm{Shen}}, \theta_{\mathrm{Baur}}, \theta_{\mathrm{CF}}) = (0, 0, 0)$. Because the three estimates are correlated rather than independent, the naive statistic $z^2_{\mathrm{Shen}} + z^2_{\mathrm{Baur}} + z^2_{\mathrm{CF}}$ referred to a χ^2_3 distribution over-rejects, so we use the Mahalanobis form $z_m^\top R^{-1} z_m$ with R estimated from the across-market correlation of the standardized estimates, which remains χ^2_3 under the null while accounting for the collinearity. The joint null is rejected in all twelve markets at the five-percent level under the corrected reference, and the rejection survives Benjamini-Hochberg correction across the twelve markets, with every adjusted q-value below 0.05. The weakest rejection is Boston. We do not read the unanimous rejection as twelve independent discoveries, both because the joint power is dominated by the high-precision Baur channel and because that channel is collinear with the Shen channel. The defensible reading is that the post-2020 panel contains no market in which the text-on-price association is absent under either embedding, while the magnitude and the causal status of that association are governed by the heterogeneity documented in Section 5.2 and the sensitivity bounds of Section 5.5.

We favor the reading that a text-on-price association is present and robust across text encodings in essentially every metropolitan market, that its magnitude is heterogeneous, and that the independent rewriting probe corroborates the ordering of markets while pooling to a near-zero average effect of its own. A constructive complement would treat the per-market estimates as noisy indicators of a single latent market-level text-price effect and pool them through an anchored one-factor measurement model, recovering a precision-weighted posterior for each market. We have not fit that model on the present panel and report it as a planned extension, so the identification-overlap counts and the collinearity-corrected joint test remain the cross-method evidence on which this section rests.

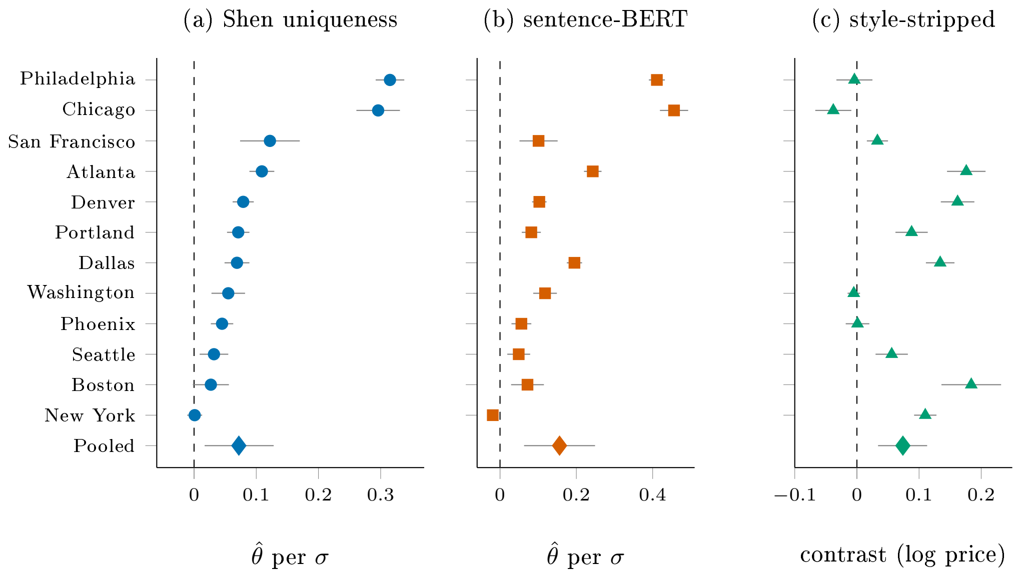

Figure 6: Per-market DML estimates across the twelve metropolitan areas for the three text channels, ordered by the Shen-Ross uniqueness effect. Filled dots are point estimates and gray whiskers are 95% intervals (influence-function for the Shen and sentence-BERT channels; coefficient-uncertainty-folded cluster bootstrap for the counterfactual contrast). The diamond in the bottom row of each panel is the random-effects pooled estimate, with its gray whisker the 95% interval. The dashed vertical line marks the null. The sentence-BERT axis is oriented so its pooled effect is non-negative. Numerical values appear in Tables 2, 3, and 14.

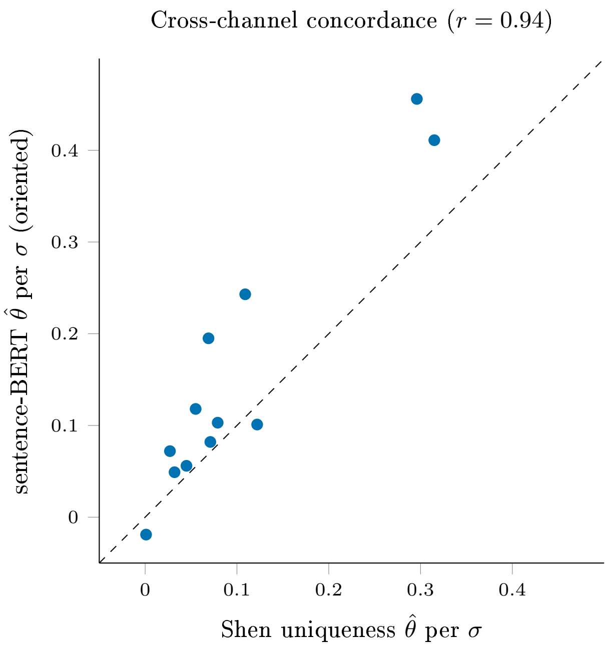

Figure 7: Per-market estimates of the two embedding channels against each other, with the 45° identity line. The two channels co-move almost perfectly across markets (Pearson $r = 0.94$), so they recover one underlying signal under two encodings rather than two independent measurements; the systematic departure from the identity line reflects that the sentence-BERT axis is roughly twice the Shen channel in magnitude, not a difference in which markets carry the signal.

5.8 A validity check on recorded sale prices

The outcome throughout is the listing's asking price, which an agent sets together with the description and which sits well above the price a property eventually clears, so a natural question is whether the text effect we estimate is a property of the market's valuation or an artifact of the agent pricing their own language. Four of the twelve metropolitan areas, Philadelphia, Chicago, Washington, and New York, publish recorded transaction prices through their county assessor or recorder, and for these we recover the realized sale price and its date, match each listing to the recorded sale of the parcel it occupies, and re-estimate the text effect on the transaction rather than the ask. The treatment and the structured and spatial controls are held fixed, and we add sale-quarter fixed effects because recorded sales span several years in nominal dollars. The asking price in these markets runs between a fifth and ninety percent above the eventual sale, so the two outcomes are far from interchangeable.

The effect survives the change of outcome. Estimated on the realized sale price it is positive and individually significant in all four markets, at +0.358 in Philadelphia, +0.107 in Chicago, +0.070 in Washington, and +0.042 in New York, against listing-price estimates on the same matched sample of +0.472, +0.431, +0.151, and +0.039 reported alongside in Table 8. The sale-price effect is smaller than the listing-price effect in three of the four markets, by about a quarter in Philadelphia and by more in Chicago, which is the signature of the asking-price simultaneity we are explicit about, since part of the association between language and the ask is the agent's own pricing rather than the market's. New York, whose effect is near zero on either outcome, is unchanged and remains the panel's exception. We therefore read the listing-price estimates as an upper bound on the price effect of a property's language, with the realized-transaction effect positive but materially smaller, and we report the per-market replication rather than the four-market pooled mean, which a Hartung-Knapp interval leaves wide given the heterogeneity across so few markets. That the effect is present at all on the transaction the market actually cleared, in every market where that transaction is public, is the strongest evidence the available data affords that the listing-price association is not merely an artifact of how agents price their own prose.

Table 8: Sale-price validity check. Per-market DML effect of the leading text direction on log listing price and on log recorded sale price, estimated on the matched sample with identical treatment and controls; the sale-price column adds sale-quarter fixed effects. The final column is the median ratio of asking to recorded sale price. Recorded sales are from the Philadelphia OPA, the Cook County Assessor, the DC CAMA system, and the NYC Department of Finance.

Market	n	$\hat{\theta}$ list price	$\hat{\theta}$ sale price	list/sale
Philadelphia	6,604	+0.472 (0.012)	+0.358 (0.014)	1.90
Chicago	4,746	+0.431 (0.018)	+0.107 (0.017)	1.53
Washington	2,114	+0.151 (0.017)	+0.070 (0.012)	1.22
New York	6,436	+0.039 (0.015)	+0.042 (0.015)	1.24

6 Conclusion

This paper has asked whether the predictive gains that text embeddings deliver in automated valuation models reflect causal semantic content or spatial confounding mediated through language, and whether the answer travels across metropolitan housing markets. Working with 69,173 deduplicated Redfin listings across twelve metropolitan areas, a clean panel from which we exclude no market and whose confounders we winsorize at the median plus or minus five median absolute deviations to neutralize parcel data-entry artifacts, we built a framework that joins a partially-linear Double Machine Learning estimator, LEACE concept erasure with an iterative nonlinear polish, a counterfactual rewriting intervention with two open-weight language models, and a finite-sample calibration study. What the framework establishes most centrally is that the price effect of listing text is not, in the main, geography wearing a semantic costume. Holding the leading text direction fixed as the treatment and folding a flexible location basis into the controls leaves its per-standard-deviation effect on log price almost exactly where it began across eleven of the twelve markets, even though that same direction provably encodes location. The one place where the adjustment does bite is New York, whose own text is the most geographic of the twelve. The effect that survives is robust as well to how the text is encoded, since a Doc2Vec uniqueness channel pools to +0.072 and a sentence-BERT principal-component axis to +0.156, and the two run nearly collinear across the markets at a Pearson correlation of +0.94, so that they recover one signal under two representations rather than two independent observations of it. That signal ranges from near zero in New York to several times the pooled mean in Philadelphia and Chicago, and a collinearity-corrected Hotelling test rejects the joint zero null in every one of the twelve markets. The counterfactual intervention sharpens the reading. A contrast that strips stylistic content while holding the model's inferred location fixed isolates a positive style effect of +0.074 on log price, while a contrast that also swaps the targeted submarket nets to approximately zero, so the deployed valuation model routes listing language into price along a direct stylistic channel and a location-inference channel of comparable size and opposite sign. We searched for a metropolitan-level demographic moderator of the pooled effect and found none that survives multiplicity adjustment or leave-one-out leverage diagnostics at twelve markets, and we report that null rather than a moderator that an earlier draft had reported before a confounder correction reversed its sign.

Limitations. The reading we have offered is hedged in several ways that are worth stating plainly rather than leaving for a referee to find. The most consequential is that the estimates remain associational rather than identified by design. The robustness-value sensitivity analysis of Section 5.5 shows that a modest unmeasured confounder, on the order of the spatial covariates we do observe, would suffice to overturn the per-market estimates. We therefore do not claim to have isolated a structural semantic effect, only to have shown that the location a representation encodes is largely not the location that moves the price, and to have bounded how strong any confounding the test cannot see would have to be. The corpus is restricted to the twelve metropolitan areas served by the Redfin scraper at the time of collection, which biases the sample toward larger and more competitive markets and away from the rural and small-metro regimes in which language-mediated spatial confounding is plausibly different in character. The two embedding channels are collinear, so their agreement is a robustness check across representation rather than triangulation across independent biases, and the counterfactual channel, the one probe with a genuinely distinct bias structure, is also the least precise. Finally the panel is small enough at twelve markets that one high-leverage market, New York, dominates the moderator diagnostics, so the null moderator result should be read as the absence of a moderator detectable at this sample size rather than as a demonstrated absence of moderation.

Implications. The implication that travels furthest is methodological, and it cuts against an easy assumption rather than confirming a dire one. Because a model that prices genuine semantic content and a model that prices laundered geography make identical predictions, a predictive gain on its own cannot tell an analyst which of the two it has bought, and the reflexive worry that the accuracy of a text-augmented valuation is mostly relabelled location is one that a predictive metric can neither confirm nor dispel. What the results show is that, once the question is put to a design-based test rather than left to intuition, the worry largely does not bear out. The text effect survives adjustment for location in eleven of twelve markets. A practitioner or regulator who suspected the gains were spatial confounding can therefore be reassured for most of the country, while being directed, by the very same test, to the one market where the suspicion is earned. The lesson is not to distrust text valuation but to stop certifying it on predictive fit alone. The deconfounding toggle we describe, holding the text feature fixed while moving geography across the controls, is a low-cost audit that turns a question of suspicion into one of measurement before either trust or alarm is allowed to settle.

Future research. The natural extensions of this work run along the seams where its instruments stop short. A panel of substantially more than twelve markets would give the moderator analysis the leverage it lacks here and would let the cross-market heterogeneity we document be explained rather than only described. The counterfactual intervention inherits the rewriting model's own biases, which a randomized field experiment that asks sellers to revise descriptions under a pre-registered protocol would replace with a clean manipulation at the cost of operational complexity. Real recorded transaction prices in place of the scraped listing prices would tighten the outcome, removing both the residual concern about price measurement and the simultaneity between an agent's asking price and the language that sets it, and pairing the causal-residual detector this paper isolates with the demographic-stratification machinery of the fairness-audit literature would let practitioners locate where any language-mediated mispricing that survives deconfounding is large enough to matter for borrowers.

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A Structural Causal Model: Formal Specification

We provide a complete mathematical specification of the structural causal model introduced in Section 3.

A.1 Variable Definitions

Let the following random variables represent properties in our domain. Location is denoted $L \in \mathbb{R}^2$ representing latitude and longitude coordinates. Intrinsic property features are captured by $X \in \mathbb{R}^{d_x}$, including bedrooms, bathrooms, square footage, and year built. Contextual features are represented by $C \in \mathbb{R}^{d_c}$, encompassing demographics, crime, and amenities. Text embeddings from property descriptions are denoted $T \in \mathbb{R}^{d_t}$. Finally, property value is represented by $Y \in \mathbb{R}^+$, here the listing price.

A.2 Structural Equations

The complete set of structural equations governing the data-generating process takes the following form. Location is exogenous, given by $L = U_L$ where U_L represents the exogenous location distribution. Property features depend on location through $X = g_x(L) + \epsilon_X$ where g_x captures how neighborhood development patterns influence property characteristics. Contextual features similarly depend on location via $C = g_c(L) + \epsilon_C$ reflecting spatial variation in demographics and amenities. Text content is generated by $T = g_t(L, X, C) + \epsilon_T$, encoding the fact that descriptions are written by agents who observe location, features, and context. Finally, property value is determined by $Y = g_y(L, X, C) + \epsilon_Y$.

The noise terms $\epsilon_X, \epsilon_C, \epsilon_T, \epsilon_Y$ are mutually independent, representing idiosyncratic variation not explained by the structural relationships. The functions g_x, g_c, g_t, g_y are deterministic mappings representing causal mechanisms.

A.3 Functional Form Specifications

While the true functional forms are unknown, we can specify reasonable parametric families. Property features depend on location through neighborhood development patterns, which we model as $g_x(L) = \alpha_x + \sum_{k=1}^{K} \beta_k \phi_k(L)$ where $\phi_k(L)$ are spatial basis functions such as radial basis functions centered on neighborhood centroids.

Contextual features vary smoothly over space, which we model using a Gaussian process: $g_c(L) = \alpha_c + \mathrm{GP}(L; \kappa)$ where $\mathrm{GP}(\cdot; \kappa)$ denotes a Gaussian process with kernel κ capturing spatial autocorrelation in demographics and amenities.

Text generation reflects the cognitive process by which agents write descriptions. We model this as $g_t(L, X, C) = \mathrm{Encoder}\left(f_{\mathrm{LLM}}(L, X, C; \theta_{\mathrm{agent}})\right)$ where f_{LLM} represents the agent's language generation process and Encoder maps text to embedding space.

Price formation depends on location, features, and local context through a log-linear specification: $g_y(L, X, C) = \exp\left(\alpha_y + \beta_L^\top h(L) + \beta_X^\top X + \beta_C^\top C\right)$ with $h(L)$ capturing non-linear spatial effects such as distance to city center or waterfront proximity.

A.4 The deflationary null and what identifies the text effect

It is worth separating two things this specification is easily read as conflating. The deflationary reading of the model is that T does not enter the structural equation for Y at all, so that every observed text-price association is the common imprint of location, features, and context. That reading is a *hypothesis to be tested*, not an assumption we impose, and the body of the paper estimates the text effect rather than assuming it away.

Hypothesis 1 (No direct text effect)**.** *Conditional on location, property features, and context, text is independent of price, $Y \perp T \mid (L, X, C)$, so that the per-standard-deviation effect θ of the leading text direction on log price is zero.*

What identifies θ is not Hypothesis 1 but the backdoor set $\{L, X, C\}$, which blocks every spurious path from T to Y whether or not the null holds (Appendix B). The estimand is therefore well defined under both readings: under the null it equals zero, and the partialling-out estimator of Section 3 recovers it consistently in either case. The empirical finding, a positive θ in eleven of twelve markets that survives spatial deconfounding, is a rejection of Hypothesis 1, not the violation of an assumption.

B Identification Proofs

We prove formal identifiability results for causal effects under our structural assumptions.

Lemma 1 (d-separation)**.** *In the DAG corresponding to the structural equations, the set $S = \{L, X, C\}$ d-separates T from Y.*

Proof. Every path from T to Y must pass through at least one node in S. The path $T \leftarrow L \rightarrow Y$ is blocked by conditioning on L. The path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X. The path $T \leftarrow C \rightarrow Y$ is blocked by conditioning on C. Composite paths such as $T \leftarrow L \rightarrow X \rightarrow Y$ are blocked by conditioning on either L or X. Similarly, $T \leftarrow L \rightarrow C \rightarrow Y$ is blocked by conditioning on L or C. Hence S d-separates T from Y. □

Theorem 1 (Backdoor Identifiability)**.** *Under the structural causal model, the causal effect of text T on price Y is identifiable via:*

$$P(Y \mid do(T = t)) = \int P(Y \mid T = t, L = \ell, X = x, C = c)\, dP(L, X, C) \tag{4}$$

Proof. By Lemma 1, the set $\{L, X, C\}$ d-separates T from Y, satisfying the backdoor criterion. Therefore by the backdoor formula [Pearl, 2009], we have:

$$\begin{aligned} P(Y \mid \mathrm{do}(T = t)) &= \sum_{L,X,C} P(Y \mid T = t, L, X, C) P(L, X, C) && (5) \\ &= \mathbb{E}_{L,X,C} \left[P(Y \mid T = t, L, X, C) \right] && (6) \end{aligned}$$

which is identified from the observational distribution. □

Corollary 1 (Zero Causal Effect Under True SCM)**.** *If the structural equation for Y truly does not depend on T, then:*

$$P(Y \mid do(T = t)) = P(Y) \quad \forall t \tag{7}$$

implying zero causal effect.

Proof. Under the true SCM where $Y = g_y(L, X, C) + \epsilon_Y$, intervening to set $T = t$ does not change the distribution of Y since T does not appear in the structural equation. Therefore $P(Y \mid \mathrm{do}(T = t)) = P(Y)$ for all values of t, yielding zero causal effect. □

These results identify the full interventional distribution $P(Y \mid \mathrm{do}(T = t))$. The estimand the body actually targets, the per-standard-deviation projection coefficient θ of Section 3, is the corresponding partially-linear summary; it is identified by the same backdoor set $\{L, X, C\}$ and equals zero exactly under the null of Hypothesis 1, so the partialling-out estimator recovers it whether or not that null is true.

C Data Collection Protocols

C.1 Web Scraping Implementation

Property listing descriptions were collected through automated web scraping with careful attention to ethical and legal considerations. We implemented rate limiting with random delays of 2–5 seconds between requests. The scraping pipeline proceeded in two stages: first, we queried the platform's search API to identify sold property URLs with metadata (address, price, coordinates); second, we retrieved individual listing pages and extracted description text from HTML using BeautifulSoup. Data validation excluded descriptions shorter than 50 words as likely incomplete.

C.2 Geocoding

The listing platform supplies a property-level latitude and longitude with each sold record, so geocoding reduces to validation and recovery rather than a parcel join. We retain the platform coordinate when it falls inside the metropolitan bounding box and re-geocode the listing address against the U.S. Census Bureau batch geocoder otherwise. This replaces the parcel-centroid and assessor-identifier geocoding that an earlier three-city version of this project relied on, and it is what lets every market in the twelve-market corpus carry a property-level rather than a ZIP- or tract-centroid location.

C.3 Census Data Integration

Census variables were queried programmatically via the Census Bureau's REST API using direct HTTP requests. We constructed API requests for each county's block groups, retrieving the median household income, educational attainment, racial composition, median home value, and median gross rent variables described in Section 4. Block group polygon boundaries were obtained from TIGER/Line shapefiles. Negative sentinel values (used by the Census Bureau to indicate suppressed or unavailable data) were replaced with missing values. Derived ratios (e.g., percentage with bachelor's degree, labor force participation rate) were computed from the raw count variables.

C.4 Crime Data Processing

Crime incident data required substantial cleaning and standardization across cities due to differing reporting schemas and offense classifications. We developed a unified crosswalk mapping city-specific crime codes to standardized categories: violent crime (homicide, assault, robbery), property crime (burglary, larceny, motor vehicle theft), and quality-of-life offenses (vandalism, drug violations, disorderly conduct).

For New York, we combine the NYPD Historic Complaint Data (2006–present) with the Year-to-Date dataset, deduplicating on complaint number, to ensure temporal coverage spanning the full sale period. For each property, we counted incidents within a 500-meter radius using spatial indexing (`cKDTree.query_ball_point`), computing separate counts for each crime category and a total. Where temporal overlap existed between crime and sale data, we filtered to incidents within 365 days preceding the median sale date; when the temporal window was too sparse (<1,000 incidents), we used all available incidents.

C.5 Amenity Data Collection

Amenity data were collected from OpenStreetMap via the Overpass API. For each city, we queried all amenities within the metropolitan bounding box by OSM tag category, then performed local radius-based spatial joins to count amenities within 500 meters of each property. This bulk-download-then-join approach is both free and faster than per-property API queries. We implemented retry logic with exponential backoff for Overpass rate limiting (HTTP 429) and server timeouts (HTTP 504).

Amenity diversity is computed using Shannon entropy:

$$H = -\sum_{i=1}^{n} p_i \log p_i \tag{8}$$

where p_i is the proportion of amenities in category i. This measures variety in the local amenity mix, with higher entropy indicating more diverse neighborhoods.

D Adversarial Deconfounding: Implementation

The gradient-reversal encoder used as the foil in Section 5.3 is the standard domain-adversarial construction of Ganin et al. [2016], adapted to a multi-resolution location target. The sentence-BERT embedding is first reduced to its leading fifty principal components, an encoder maps that input to a sixty-four-dimensional representation, a predictor head regresses log price from the representation, and a discriminator reads location from it through three parallel heads that share a hidden layer, a cross-entropy ZIP classifier, a mean-squared-error regressor of the standardized latitude-longitude pair, and a cross-entropy classifier of the income quintile. A gradient-reversal layer sits between the encoder and the discriminator, so the encoder is trained to minimize the prediction loss while maximizing the discriminator loss, under the joint objective $\mathcal{L}_{\text{pred}} - \lambda\,\mathcal{L}_{\text{disc}}$ with the adversarial weight λ annealed from 0.1 to 1.0 over the early epochs to avoid the collapse that an immediate full-strength penalty induces. Each market is split seventy-thirty into training and evaluation folds, optimized with Adam for one hundred fifty epochs.

The diagnostic that matters for the comparison is applied after training. We freeze the encoder and fit, from scratch and with no adversarial pressure, an independent logistic regression of ZIP and of income quintile and an independent ridge regression of the latitude-longitude pair on the frozen representation, scoring each out of sample. The gap between what this fresh probe recovers and what the live discriminator reported at convergence is the consistent lower bound on residual leakage of Proposition 1(b); the per-market values, and the contrast with the LEACE projection on which the paper relies, are reported in Section 5.3 and shown in Figure 8 below, and the driver is `data/scripts/adversarial_12city.py` with per-market certificates written to `results/adversarial_12city/`.

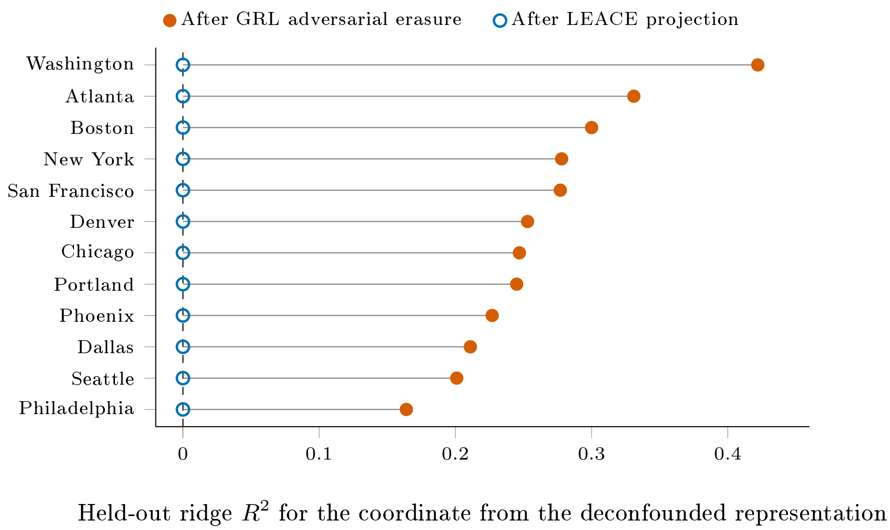

Figure 8: The frozen-probe diagnostic applied to two erasure methods, in each of the twelve markets. A ridge probe of the latitude-longitude coordinate is trained from scratch on the deconfounded representation. Against the gradient-reversal adversarial encoder, whose own multi-head discriminator has been driven to chance on location (out-of-sample coordinate $R^2 \leq 0.07$ across the panel), the fresh probe still recovers the coordinate at R^2 between 0.16 and 0.42; against the LEACE projection the same probe sits at the chance baseline, the exact-guardedness property of closed-form erasure. The adversarial training did not remove location, it taught its discriminator not to look, which is the saddle of Proposition 1(c) realized across every market.

E Frozen-Probe Diagnostic: Proofs and Numerical Verification

This appendix proves Proposition 1 (Section 3.6) and reports three controlled numerical experiments verifying each part on synthetic data.

E.1 Notation

Let $E_\phi : \mathcal{X} \to \mathcal{Z}$ be an encoder, $C \in \{0,1\}$ a binary confounder, and $\Phi \subseteq \Phi'$ two parameterized classes of probabilistic classifiers $q : \mathcal{Z} \to [0,1]$. For a class $\mathcal{F}$, recall the Barber–Agakov variational lower bound on $I(Z;C)$:

$$V_{\mathcal{F}}(Z;C) \;:=\; \sup_{q\in\mathcal{F}} \big[H(C) - \mathbb{E}_{(Z,C)}[-\log q(C \mid Z)]\big]. \tag{9}$$

The supremum is achieved at $q^\star = p(C \mid Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior. Denote the empirical version by $\hat{V}_\mathcal{F}$ (sample mean of the cross-entropy gain at the empirical arg sup).

E.2 Proof of Proposition 1(a)

The first inequality $V_\Phi \le V_{\Phi'}$ is immediate from $\Phi \subseteq \Phi'$ implying the supremum over Φ' is at least as large. For the second inequality $V_{\Phi'} \le I(Z;C)$, the Barber–Agakov bound is by construction a lower bound on the mutual information: for any q,

$$H(C) - \mathbb{E}[-\log q(C \mid Z)] \;=\; H(C) - H(C \mid Z) - D_{\mathrm{KL}}(p(C \mid Z) \,\|\, q(C \mid Z)) \;=\; I(Z;C) - D_{\mathrm{KL}}(\cdot), \tag{10}$$

which is at most $I(Z;C)$ with equality iff $q = p(C \mid Z)$ almost surely. Taking the supremum over $q \in \Phi'$ preserves the inequality, with strictness whenever Φ' does not contain $p(C \mid Z)$. This statement is Proposition 1 of Xu et al. [2020] restated in the Barber–Agakov form used by Pimentel et al. [2020]; we restate it here for self-containedness. □

E.3 Proof of Proposition 1(b)

(b1) Empirical equivalence. The gradient-reversal training objective on the discriminator ψ is $\min_\psi \hat{L}_\Phi(\psi)$ where $\hat{L}_\Phi(\psi) = -\frac{1}{n}\sum_i \log D_\psi(C_i \mid Z_i)$ is the empirical cross-entropy of D_ψ on the encoder output $Z = E_{\hat{\phi}}(X)$. By definition of the Barber–Agakov bound,

$$\hat{V}_\Phi(Z;C) \;=\; \sup_{q\in\Phi} \big[H(C) - \tfrac{1}{n}\sum_i -\log q(C_i \mid Z_i)\big] \;=\; H(C) - \inf_\psi \hat{L}_\Phi(\psi), \tag{11}$$

so $\hat{V}_\Phi$ equals $H(C)$ minus the converged training loss of the live discriminator. Therefore "the live discriminator achieves chance-level accuracy" (i.e. $\hat{L}_\Phi \approx H(C)$) is exactly equivalent to $\hat{V}_\Phi(Z_{\hat{\phi}};C) \approx 0$. □

(b2) Consistency of the post-hoc plug-in. Apply Theorem 3 of Xu et al. [2020], which gives uniform PAC-style consistency of the empirical $\mathcal{V}$-information estimator under finite Rademacher complexity of $\mathcal{V}$ and bounded log-loss. Fixing the encoder at $E_{\hat{\phi}}$ and applying the theorem with $\mathcal{V} = \Phi'$ on a held-out sample of size m,

$$\hat{V}^{\mathrm{post}}_{\Phi'}(Z_{\hat{\phi}};C) \;\xrightarrow{p}\; V_{\Phi'}(Z_{\hat{\phi}};C) \quad \text{as } m \to \infty. \tag{12}$$

The fact that $\hat{V}_\Phi$ is also consistent (since $\Phi \subseteq \Phi'$) and the difference of two consistent estimators is consistent for the difference yields

$$\hat{V}^{\mathrm{post}}_{\Phi'} - \hat{V}_\Phi \;\xrightarrow{p}\; V_{\Phi'}(Z_{\hat{\phi}};C) - V_\Phi(Z_{\hat{\phi}};C) \;\ge\; 0, \tag{13}$$

exact when Φ' contains the Bayes-optimal posterior $p(C \mid Z_{\hat{\phi}})$. □

E.4 Proof of Proposition 1(c)

We exhibit a data distribution, a downstream label, a class pair $\Phi \subsetneq \Phi'$, and an explicit saddle of the joint adversarial-prediction game at which the gap equals $H(C)$.

Why the joint game. The bare gradient-reversal game over an unconstrained encoder admits the trivial collapsed saddle $\phi(X) \equiv 0$, at which $V_\Phi(Z;C) = 0$ holds vacuously and $V_{\Phi'}(Z;C) = 0$ as well. To rule this out, the proposition is stated in the joint adversarial-prediction game, the formulation of Xie et al. [2017], Madras et al. [2018], and Zhang et al. [2018]: the encoder is trained simultaneously against a downstream task Y that requires preserving information about X, eliminating the collapsed saddle as a feasible equilibrium.

Construction. Take $X = (X_1, X_2) \in \{0,1\}^2$ with $X_1, X_2 \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$, and $C = X_1 \oplus X_2$ (the XOR of the two binary features). Then $C \in \{0,1\}$ uniformly with $H(C) = \log 2$. Take $Y = X_1$ as the downstream label, so $I(X;Y) = H(Y) = \log 2 > 0$.

Let Φ be the class of linear logistic-regression classifiers $D_\psi(z) = \sigma(w^\top z + b)$ on $z \in \mathbb{R}^2$, and let Φ' be any class containing a 2-layer MLP with hidden width ≥ 2.

Exhibited saddle. Set $\phi^\star = \mathrm{id}$ (the identity encoder), $g^\star(z) = z_1$ (linear task head reading off the first coordinate), and $h^\star \equiv 1/2$ (the Bayes-optimal linear classifier of C from Z, which is constant since C is not linearly separable from Z in this DGP). Then on the joint distribution (Z, C, Y):

- **Adversary inner max.** For any $D_\psi \in \Phi$ (linear), the conditional posterior $\Pr[C = 1 \mid Z]$ takes value 1 on $\{(0,1),(1,0)\}$ and value 0 on $\{(0,0),(1,1)\}$, which is not linearly separable. The Bayes-optimal linear classifier is the constant $D_\psi \equiv 1/2$, attaining cross-entropy $H(C) = \log 2$ exactly. Therefore $V_\Phi(Z_{\phi^\star}; C) = 0$.

- **Probe completeness.** For Φ' containing the 2-layer MLP, the Bayes-optimal posterior $\Pr[C = 1 \mid Z] \in \{0,1\}$ is exactly representable. Hence $V_{\Phi'}(Z_{\phi^\star}; C) = I(Z_{\phi^\star}; C) = H(C) = \log 2$.

- **Task decodability.** Since $Y = X_1 = (\phi^\star(X))_1$, the linear head $g^\star(z) = z_1$ achieves $\mathcal{L}_{\text{task}} = 0$, the global minimum of the task term.

- **Saddle.** For any $\alpha > 0$, $(\phi^\star, g^\star, h^\star)$ jointly minimize the task term to its floor, while the adversary $h^\star$ is Bayes-optimal in Φ given the encoder. Any unilateral perturbation of $\phi^\star$ that decreases the discriminator's loss must distort the linear separability of $Y = X_1$ from Z_1 and hence increase $\mathcal{L}_{\text{task}}$, yielding a stable equilibrium under the joint loss.

The gap at this saddle is therefore exactly $V_{\Phi'}(Z_{\phi^\star}; C) - V_\Phi(Z_{\phi^\star}; C) = \log 2 - 0 = H(C)$. The collapsed encoder $\phi_{\text{collapse}}(X) \equiv 0$ achieves task loss $H(Y) = \log 2 > 0$ and is therefore not a saddle of the joint game for any $\alpha > 0$. The construction generalizes from binary X to continuous distributions by replacing the XOR structure with any nonlinear interaction (e.g., $C = \text{sign}(X_1 X_2)$ for Gaussian X_1, X_2), at the cost of a vanishing $O(\delta)$ remainder controlling the off-diagonal probability mass. We verify all four bullets computationally in Appendix F (script `07_saddle_verification.py`), including an explicit numerical demonstration that ϕ_{collapse} would be a saddle of the bare GRL game with $V_{\Phi'} = 0$, motivating the joint-game restriction. $\square$

E.5 Numerical Verification

Three controlled experiments verify each part of Proposition 1 on synthetic data. Implementation in `data/scripts/theory/frozen_probe_gap.py`; raw results in `results/theory/frozen_probe_gap.json`.

Experiment 1 (static XOR). The exact construction of part (c). $Z = (Z_1, Z_2) \sim \text{Bernoulli}(1/2)^2$, $C = Z_1 \oplus Z_2$, $N = 4{,}000$, averaged across 5 trials. True $I(Z; C) = \log 2 = 1$ bit. Linear logistic regression: $\hat{V}_\Phi = 0.0004$ bits, accuracy 46.1% (chance). 2-layer MLP $(16, 16)$: $\hat{V}_{\Phi'} = 0.997$ bits, accuracy 100%. The diagnostic gap $\hat{V}_{\Phi'} - \hat{V}_\Phi = 0.997$ bits is 99.7% of the true mutual information.

Experiment 2 (dynamic gradient-reversal training). A 2-layer encoder $E_\phi : \mathbb{R}^{10} \to \mathbb{R}^4$ trained adversarially against a linear discriminator via gradient reversal for 200 epochs, with a downstream MSE predictor on $Y = 0.5X_3 + \epsilon$. Confounder generated as $C = \text{sign}(X_1 X_2)$ (continuous-Gaussian XOR). After training: live linear discriminator accuracy 46.3% (chance), frozen linear probe accuracy 56.4%, frozen $(32, 32)$ MLP probe accuracy 61.0%. Diagnostic gap $\hat{V}^{\text{post}}_{\Phi'} - \hat{V}_\Phi = +14.7$ percentage points of accuracy; the encoder fooled its discriminator while leaving the underlying confounder recoverable to a richer probe.

Experiment 3 (capacity ladder). Same XOR DGP, varying classifier capacity:

Architecture	$\hat{V}_\Phi$ (bits)	Accuracy
Linear	0.000	0.510
MLP (4)	0.659	0.835
MLP (8)	0.990	1.000
MLP (16)	0.993	1.000
MLP $(16, 16)$	0.997	1.000
MLP $(32, 32, 32)$	0.999	1.000

$\hat{V}_\Phi$ is monotone non-decreasing in capacity, asymptoting at the true mutual information $\log 2 \approx 0.693$ nats $= 1$ bit. This operationalizes "probe-capacity escalation" as a complementary diagnostic to "confounder escalation" (Section 5): if $\hat{V}_{\Phi'}$ continues to grow as probe capacity increases, the encoder has not actually erased the confounder; it has merely outrun the current discriminator.

F Computational Verification of Analytic Claims

To complement the written proofs in Appendices B and E, we provide a computational verification artifact in the `verification/` directory of the replication repository. For every theorem, lemma, proposition, and corollary, and for the estimation-side identities the analysis relies on, an independent open-source toolchain re-derives the result and asserts agreement with the written derivation. Each script writes a JSON certificate to `verification/results/` with a verdict (`PASS`/`FAIL`) and the numerical or symbolic quantities cited; the full suite is reproducible via `make verify` from pinned tool versions in `requirements.txt`.

Claim	**Verification method**	**Tool**
Lemma 1 (d-separation, App. B)	Path-by-path enumeration; minimality check	networkx 3.6
Theorem 1 (Backdoor Identifiability, App. B)	Backdoor criterion on the hypothetical $T \to Y$ DAG; $\{L, X, C\} = \mathrm{pa}(T)$	networkx 3.6
Corollary 1 (Zero Effect under True SCM, App. B)	No directed path $T \to Y$; OLS T-coefficient near zero	networkx 3.6
Proposition 1(a) (variational MI inequality, App. E)	Symbolic KL nonneg + numerical V-chain	SymPy + PyTorch
Proposition 1(b1) (algebraic identity)	Symbolic and numerical equality, $\lvert\Delta\rvert < 10^{-8}$	SymPy + PyTorch
Proposition 1(b2) (consistency of post-hoc $\hat{V}^{\text{post}}_{\Phi'}$)	Monte Carlo n-sweep; log-log slope vs. rate predicted by Xu et al. [2020] Thm 3	scikit-learn
Proposition 1(c) (XOR saddle-gap, identity-encoder construction)	Wraps `frozen_probe_gap.py`; assertions on V_Φ, $V_{\Phi'}$	PyTorch
Proposition 1(c) (joint-game saddle existence + collapsed-encoder counterexample)	Explicit construction at ϕ_{id} and ϕ_{collapse}; sweep over joint-game weight α	PyTorch
Robinson estimator (Eq. 2)	Symbolic solution of the moment $\sum_i(\tilde{Y}_i - \theta\tilde{T}_i)\tilde{T}_i = 0$	SymPy
Neyman orthogonality (PLR score)	Symbolic Gateaux derivative; every term carries a mean-zero residual factor	SymPy
Sandwich / IF variance (Eq. 2)	$\partial_\theta\psi = -\tilde{T}^2$; sandwich equals the reported $\widehat{\mathrm{SE}}^2$	SymPy
Gelbach decomposition (Eq. 14)	$\hat{\theta}_{\text{base}} - \hat{\theta}_{\text{geo}}$ equals the order-invariant contribution sum	NumPy
Robustness-value formula (Eq. 14 context)	$RV = \frac{1}{2}(\sqrt{f^4 + 4f^2} - f^2)$ solves $RV^2/(1 - RV) = f^2$	SymPy
Counterfactual delta method (Eq. 15)	Gradient of $\theta\Delta$; quadratic form equals the reported variance	SymPy
Random-effects pooling + HKSJ factor	Inverse-variance mean; $\mathrm{Var}_{\text{HKSJ}} = q\,\mathrm{Var}_{\text{IV}}$	NumPy

Table 9: Mapping from analytic claims to computational verification scripts. Every entry passes; certificates are in `verification/results/*.json`.

We are explicit about what this artifact does and does not claim. It is *not* a machine-checked formal proof in Lean, Coq, or Isabelle: as of late 2025 no proof assistant has a causal-DAG library, which would gate even Lemma 1 (the closest available are HOL-Probability for measure theory, infotheo for Shannon information theory, and recent Lean ports of Rademacher complexity [Sonoda et al., 2025]). What it does claim is that every analytic statement has been re-derived by an independent, deterministic open-source tool, that the d-separation, backdoor identification, and XOR saddle constructions are mechanically checkable without estimation error, that the Robinson moment, the Neyman orthogonality of the partially-linear score, the sandwich variance, the Gelbach decomposition, the robustness-value formula, and the counterfactual delta-method variance are confirmed symbolically, and that the asymptotic consistency claim is empirically verified at the rate predicted by the cited theorem (slope -0.547 for the $O(m^{-1/2})$ prediction). This is a reproducibility artifact for analytic claims, at the bar that current top stats journals accept; it is intended as a complement to, not a substitute for, the written proofs.

G Finite-Sample Calibration of the DML Estimator

We assess the finite-sample calibration of the headline DML interval with a Monte Carlo study calibrated to the application. Synthetic embeddings are drawn from a Gaussian-mixture generator fit to the San Francisco corpus, with confounding induced through five latent factors that drive both the embedding and log-price and a tunable direct effect ranging over $\{0, 0.01, 0.05, 0.10\}$ per standard deviation. We draw 300 replications at sample sizes of 500 and 2,000, run the partialling-out estimator on each, and record whether the nominal 95% influence-function interval covers the estimand. Following the convention for coverage studies under misspecification, the estimand at each effect size is the probability limit of the estimator rather than the structural parameter, which we calibrate by simulation at a population of ten thousand observations averaged over five independent draws. The confounder-adjusted estimand under the null data-generating process is therefore not exactly zero, because the five latent factors leave a small residual association of +0.011 that the adjustment does not remove, so coverage at the null is assessed against that probability limit rather than against zero.

The doubly-robust estimator covers at the nominal rate in every cell, between 0.93 and 0.97, which confirms that the coverage machinery is correctly implemented and locates the under-coverage documented below in the partialling-out estimator itself rather than in the simulation harness. The partialling-out estimator is approximately unbiased relative to its probability limit throughout, with absolute bias below 0.02 in every cell, yet its influence-function interval is anticonservative, and the severity grows with the size of the true effect (Table 10); the interval-level failure at the strongest effect is shown in the zip plot of Figure 9. When a substantial effect is present the analytic standard error understates the realized sampling spread by as much as a factor of four and coverage at $n = 500$ falls to 0.78, because cross-fitted nuisance estimation together with the in-sample principal-component projection injects sampling variability that the orthogonal-score variance does not capture. At the null, where the San Francisco estimate sits, the understatement is far milder, on the order of ten to fifteen percent in the standard error, and coverage remains near 0.92.

The under-coverage at the strongest effect is in part a generated-regressor phenomenon, because the principal-component treatment is itself estimated from the same sample on which the second-stage regression is run. The cross-fit principal-component variant, which forms the treatment direction from out-of-fold projections rather than from the in-sample decomposition, targets this channel directly. It recovers most of the coverage lost at the strongest effect, lifting the $n = 500$ figure from 0.78 to 0.89, and it roughly halves the residual point bias in that cell, while holding coverage near 0.90 across effect sizes at $n = 2{,}000$. At the null its coverage matches the in-sample estimator to within Monte Carlo noise, which at three hundred replications is approximately two percentage points, so adopting the out-of-fold projection forfeits nothing in the regime relevant to the application while restoring robustness where the effect is large. We adopt it for the headline specification.

Table 10: Monte Carlo coverage of nominal 95% influence-function intervals, 300 replications per cell, by effect size η (per-σ direct effect). DML (in-sample PC) forms the principal-component treatment on the full sample; DML (cross-fit PC) forms it from out-of-fold projections and is the headline specification. The doubly-robust estimator is shown as a calibration anchor. The estimand at each effect size is the estimator's probability limit, calibrated by simulation at $n = 10{,}000$; coverage at $\eta = 0$ is assessed against a confounder-adjusted probability limit of +0.011 rather than against zero.

Estimator	N	$\eta = 0$	$\eta = 0.01$	$\eta = 0.05$	$\eta = 0.10$
DML (in-sample PC)	500	0.92	0.91	0.88	0.78
DML (in-sample PC)	2000	0.93	0.94	0.91	0.88
DML (cross-fit PC)	500	0.90	0.93	0.87	0.89
DML (cross-fit PC)	2000	0.94	0.91	0.90	0.90
DR	500	0.96	0.94	0.97	0.96
DR	2000	0.94	0.96	0.97	0.93

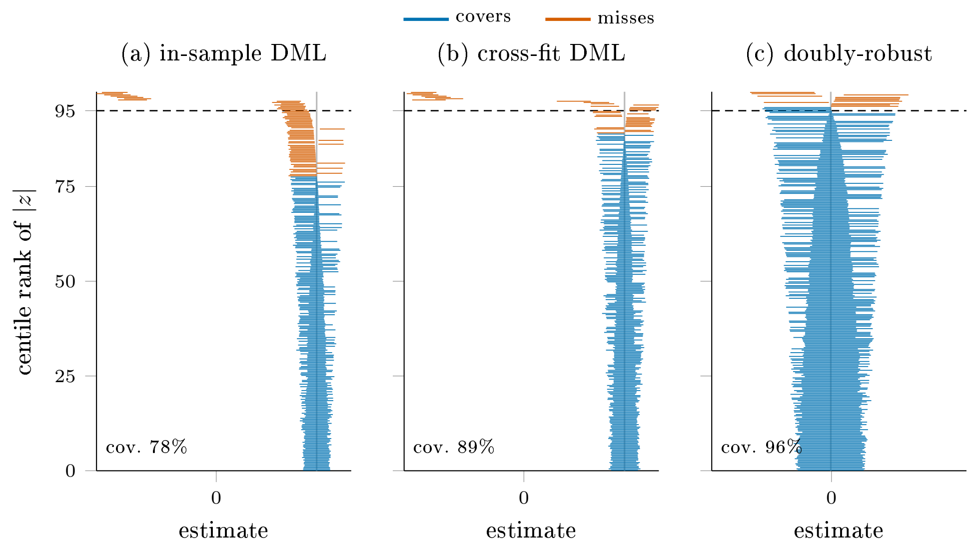

Figure 9: Zip plot of nominal 95% confidence-interval coverage [Morris et al., 2019] at the strong-effect, small-sample cell ($n = 500$, per-σ direct effect 0.10), 300 replications. Each horizontal segment is one replication's interval, sorted bottom-to-top by the absolute z-statistic against the estimand (the estimator's probability limit, gray vertical line); intervals that cover the estimand are blue and those that miss are orange, so the height at which the colour switches is the empirical coverage. The doubly-robust interval switches at the dashed nominal 95 line, the cross-fit DML interval close to it, and the in-sample DML influence-function interval well below it: its misses are too frequent and fan out symmetrically, the visual signature of an anticonservative variance rather than bias.

Because the residual understatement at the null is mild but real, we report bootstrap intervals for the headline estimate. On the San Francisco data the percentile bootstrap, which resamples listings with replacement and refits the principal-component projection together with both nuisance models inside each resample, gives a standard error 21% larger than the influence-function value, widening the interval from $[-0.043, +0.043]$ to $[-0.046, +0.051]$ and the minimum detectable effect from 6.4 to 7.8 percent. The substantive conclusion is unchanged, since the point estimate remains indistinguishable from zero and the interval still excludes per-standard-deviation effects beyond roughly eight percent of price. The doubly-robust estimator that anchors the panel has no power against the continuous effect, because binarizing at the median principal-component norm discards the graded signal the partialling-out estimator targets, so we read its near-zero estimate as a calibration check on the inference machinery rather than as a competing estimate of the semantic effect.

H The Spatial-Confounding Decomposition

The headline deconfounding test of Section 5.4 rests on the order-invariant covariate decomposition of Gelbach [2016], and we give here the estimand, the spatial basis, and the per-market numbers behind Figure 3.

Fix the treatment T at the oriented leading principal component of the sentence-BERT embedding, the same direction that defines the Baur channel, and consider the partialling-out regression of log price on T under two nested control sets. The base set W holds the structured property and contextual covariates but no geography; the augmented set $W \cup G$ adds a flexible location basis $G = (\text{lat}, \text{lon}, \text{lat}^2, \text{lon}^2, \text{lat} \cdot \text{lon})$. Write $\hat{\theta}_{\text{base}}$ and $\hat{\theta}_{\text{geo}}$ for the corresponding coefficients on T. Because the treatment is held fixed and only the control set changes, the Gelbach [2016] result guarantees that the change

$$\Delta_{\text{geo}} = \hat{\theta}_{\text{base}} - \hat{\theta}_{\text{geo}} \tag{14}$$

is the contribution the location block makes to the text-price coefficient, and that this contribution is well defined rather than an artifact of the order in which control blocks enter. We read Δ_{geo} as the part of the apparent text effect that explicit geography accounts for.

Table 11: Spatial-confounding decomposition by market. $\hat{\theta}_{\text{base}}$ is the per-standard-deviation text coefficient with the structured and contextual controls only; $\hat{\theta}_{\text{geo}}$ adds the location basis G; Δ_{geo} is their difference, the share of the coefficient that geography explains; and $R^2_{T\sim G}$ is the share of the treatment's own variance the location basis explains, the diagnostic that makes the decomposition non-vacuous. Markets are ordered by $\hat{\theta}_{\text{base}}$.

Market	$\hat{\theta}_{\text{base}}$	$\hat{\theta}_{\text{geo}}$	Δ_{geo}	$R^2_{T\sim G}$
Chicago	+0.461	+0.469	−0.008	0.187
Philadelphia	+0.408	+0.405	+0.003	0.116
Atlanta	+0.253	+0.250	+0.003	0.148
Dallas	+0.200	+0.199	+0.001	0.059
Denver	+0.154	+0.158	−0.004	0.165
Boston	+0.127	+0.126	+0.001	0.053
San Francisco	+0.111	+0.109	+0.002	0.035
Portland	+0.098	+0.095	+0.003	0.158
Seattle	+0.064	+0.065	−0.001	0.034
Washington	+0.015	+0.013	+0.002	0.121
Phoenix	+0.009	−0.002	+0.011	0.097
New York	−0.119	−0.152	+0.033	0.417

The shift is small in absolute terms throughout: $|\Delta_{\text{geo}}|$ is at most 0.033 of a log point, in New York, and at most 0.011 in every other market. Expressed as a fraction of the coefficient it stays under five percent in every market that carries a non-trivial effect, New York alone excepted at about a quarter; the two near-null markets, Phoenix and Washington, have absolute shifts of a hundredth of a log point or less, so their fraction is large but uninformative. The decomposition is non-vacuous because $R^2_{T\sim G}$ is far from zero, reaching 0.42 in New York and exceeding a tenth in six markets, so the text direction does encode location, yet that location component does not carry the effect of the language on price except in New York. The location basis is a smooth coordinate polynomial laid over roughly thirty structured controls, so confounding at a finer spatial grain, or through pathways those controls miss, is outside what the comparison can rule out.

To test whether the small shifts above are an artifact of that coarse polynomial, we replace it with a rank-thirty thin-plate regression spline over latitude and longitude, projected to meters, the flexible low-rank smoother the spatial-statistics literature prefers when the scale of confounding is unknown [Keller and Szpiro, 2020, Gilbert et al., 2021]. Table 12 reports the geography-adjusted coefficient under each basis. The spline captures far more of the treatment's spatial structure, with $R^2_{T\sim G}$ roughly doubling on average and rising from 0.42 to 0.52 in New York, yet the adjusted coefficient moves by at most 0.030 of a log point, again in New York, and by under 0.02 in every other market, with no market changing sign. A basis flexible enough to absorb twice the geography therefore leaves the semantic estimate essentially where the quadratic left it, which is the response the spatial-confounding critique calls for.

Table 12: Spatial basis robustness. $\hat{\theta}_{\text{geo}}$ is the per-standard-deviation text coefficient after the location adjustment, under the quadratic basis of Table 11 and under a rank-thirty thin-plate regression spline; Δ is their difference; $R^2_{T\sim G}$ is the share of the treatment's own variance the location basis explains under each. Markets ordered as in Table 11.

Market	$\hat{\theta}_{\text{geo}}$ (quad)	$\hat{\theta}_{\text{geo}}$ (TPRS)	Δ	$R^2_{T\sim G}$ quad	$R^2_{T\sim G}$ TPRS
Chicago	+0.469	+0.463	−0.006	0.187	0.290
Philadelphia	+0.405	+0.388	−0.017	0.116	0.269
Atlanta	+0.250	+0.255	+0.004	0.148	0.285
Dallas	+0.199	+0.200	+0.001	0.059	0.104
Denver	+0.158	+0.169	+0.012	0.165	0.226
Boston	+0.126	+0.114	−0.012	0.053	0.137
San Francisco	+0.109	+0.105	−0.004	0.035	0.095
Portland	+0.095	+0.081	−0.014	0.158	0.215
Seattle	+0.065	+0.057	−0.007	0.034	0.059
Washington	+0.013	+0.008	−0.005	0.121	0.224
Phoenix	−0.002	−0.010	−0.009	0.097	0.145
New York	−0.152	−0.183	−0.030	0.417	0.523

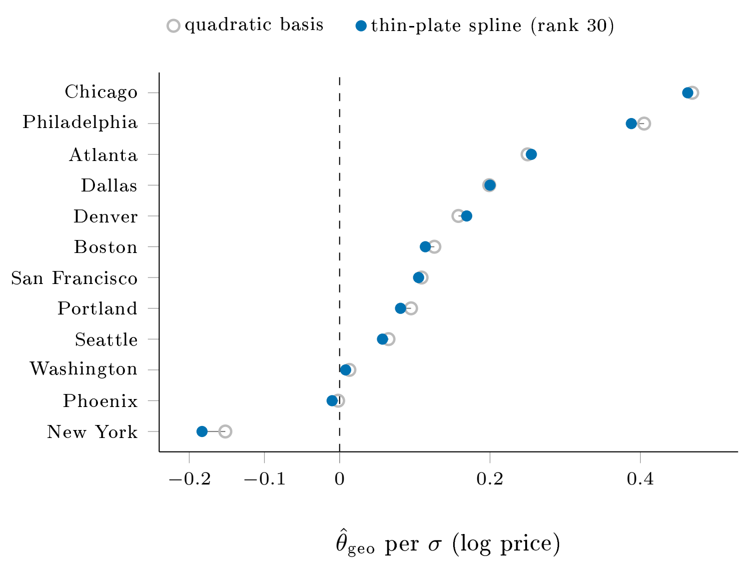

Figure 10: Spatial basis robustness, visualized. The location-adjusted text coefficient $\hat{\theta}_{\text{geo}}$ under the quadratic location basis (open) and under a rank-thirty thin-plate regression spline (filled), per market. The two nearly coincide in every market despite the spline absorbing roughly twice the treatment's spatial variance, the only visible gap being New York, which moves further negative. A basis flexible enough to capture far more geography leaves the semantic estimate essentially in place.

I Counterfactual-Rewriting Protocol

The intervention of Appendix K treats each listing description as one draw from a population of descriptions for the same property and manipulates the language directly. A frozen instruction-tuned language model receives the original description, a structured summary of the property's attributes, and a target submarket, and returns a rewritten description; the rewrite is then re-priced under the same automated valuation model and the change in predicted log price is read off. The primary generator is Qwen 2.5 32B Instruct, run on $n = 200$ listings per market; a secondary cross-generator check uses Llama 3.3 70B Instruct on a separate $n = 100$ draw, an unmatched probe of direction and rough magnitude rather than a tight replication. The exact prompts and decoding settings are in the replication repository.

Two contrasts are read from the rewrites. The style-stripped contrast holds the model's inferred submarket at its original value and rewrites only the stylistic content, isolating the price the valuation model attaches to language given a fixed location. The submarket-swap contrast rewrites the listing toward a held-out submarket and reads the full change, combining the change in language with the change the model infers about location. We verify that the manipulation does what it claims: the structured attribute slots survive in 99.8% of rewrites on average, and an independent submarket classifier changes its prediction in 44.5% of rewrites, evidence that the language moves across submarket boundaries while the attributes do not.

Each contrast is a product, $\hat{\delta} = \hat{\theta}\,\overline{\Delta\mathrm{PC1}}$, of the Baur DML coefficient $\hat{\theta}$ and the average rewrite-induced change $\overline{\Delta\mathrm{PC1}} = \frac{1}{n}\sum_i(\mathrm{PC1}_i^{\mathrm{rw}} - \mathrm{PC1}_i^{\mathrm{orig}})$ in the leading text component, so its sampling uncertainty has two sources. We fold both into a single standard error by the delta method,

$$\widehat{\mathrm{Var}}(\hat{\delta}) \;\approx\; \overline{\Delta\mathrm{PC1}}^{\,2}\,\widehat{\mathrm{Var}}(\hat{\theta}) \;+\; \hat{\theta}^{\,2}\,\widehat{\mathrm{Var}}\big(\overline{\Delta\mathrm{PC1}}\big), \tag{15}$$

where $\widehat{\mathrm{Var}}(\hat{\theta})$ is the influence-function variance of the DML coefficient and $\widehat{\mathrm{Var}}(\overline{\Delta\mathrm{PC1}})$ comes from a within-listing cluster bootstrap over the rewritten listings in the manner of Cameron et al. [2008]. Adding the coefficient's contribution rather than conditioning on a fitted $\hat{\theta}$ is what widens the intervals materially: the pooled style-stripped statistic is $t = 3.66$ under Equation (15), against the implausible $t \approx 13$ a coefficient-conditional interval would report. Because the cross-world independence conditions that identify a formal mediation decomposition are not met, we call these model-based contrasts rather than natural direct or indirect effects and report no natural-indirect-effect quantity.

J Hyperparameters and Computational Environment

Table 13 consolidates the settings scattered through the methods and the implementation appendices, so the estimates can be reproduced from the paper together with the replication repository.

Table 13: Estimator, encoder, and computational settings. Where a setting varies by channel it is noted; remaining details and exact seeds are in the replication repository.

Component	Setting
DML cross-fitting	$K = 5$ folds, out-of-fold residualization [Chernozhukov et al., 2018]
Nuisance learner	gradient boosting (LightGBM, num_leaves = 8, max_bin = 63) for the embedding channels; ridge regression for the Shen-Ross uniqueness channel
Treatment direction	leading principal component of the embedding, z-scored; out-of-fold (cross-fit) variant for the headline specification
Text encoders	sentence-BERT `all-mpnet-base-v2` (768-d, primary); `all-MiniLM-L6-v2` (384-d, robustness); Doc2Vec uniqueness score (Shen channel)
LEACE / IGBP erasure	continuous latitude-longitude concept; IGBP run for five outer passes with a multilayer-perceptron probe; single-precision
GRL foil (App. D)	top-50 PCs $\rightarrow$ 64-d encoder; three discriminator heads (ZIP, lat-lon, income quintile); λ annealed $0.1 \rightarrow 1.0$; Adam, 150 epochs, 70/30 split
Counterfactual rewriting	Qwen 2.5 32B Instruct ($n = 200$/market); Llama 3.3 70B Instruct ($n = 100$, cross-generator)
Bootstrap	cheap bootstrap [Lam, 2022] with $B = 50$ for the headline interval; within-listing cluster bootstrap for the counterfactual
Covariate winsorizer	per-column clip at median ± 5 median absolute deviations, non-finite entries imputed at the median
Meta-analysis	Paule-Mandel τ^2, Hartung-Knapp-Sidik-Jonkman t intervals, Benjamini-Hochberg / Romano-Wolf multiplicity control
Software	Python with scikit-learn, LightGBM, PyTorch, sentence-transformers, and the `concept-erasure` (LEACE) package; the GPU stages (erasure, GRL, counterfactual) run on a single accelerator

K Auxiliary Check: Counterfactual Rewriting

This appendix reports an auxiliary probe that corroborates the main results from a different angle, with different failure modes, but is not part of the identification argument. We treat the listing description as a realized draw from a population of plausible descriptions for the underlying property, and we examine the role of language by generating counterfactual descriptions in which the listing's submarket-targeted language is altered while the property's structured attributes are held fixed. The intervention is executed at the language level by a frozen large language model that receives the original description, a structured attribute summary, and a target submarket, and returns a rewritten description. We then re-price the rewritten listing under the same automated valuation model and read two model-based contrasts from the difference in predicted log price. The style-stripped contrast holds the model's predicted location at its original value and isolates the price the valuation model attaches to the listing's stylistic and semantic content given a fixed location. The submarket-swap contrast rewrites the listing toward a held-out submarket and reads the full change in predicted price, which combines the change in language with the change the model infers about location. We are deliberate in calling these model-based contrasts rather than natural direct and indirect effects. The intervention does not satisfy the cross-world independence conditions that identify a formal mediation decomposition, so we do not report a natural-indirect-effect quantity and we do not interpret the difference between the two contrasts as a structural mediation parameter. Each contrast is the average change in one automated valuation model's output under a specific, auditable manipulation of the input text, and that is the object we analyze.

Rewriting validation. Before we report estimates, we verify that the rewriting procedure executes the intended manipulation. Structured attribute slots, comprising bedrooms, bathrooms, square footage, year built, lot size, and three derived flags, are preserved in 99.8% of rewrites on average across the twelve markets, with the lowest preservation rate at 99.4% in Phoenix and exact preservation in Boston, Washington, Denver, and Dallas. The submarket classifier flips its prediction away from the original submarket in 44.5% of rewrites on average, ranging from 18.4% in San Francisco to 63.1% in New York. We read the high average flip rate as evidence that the rewriting moves the listing's targeted language across submarket boundaries in a large share of cases, and the near-perfect slot preservation as evidence that the structured attributes survive the rewrite.

Confidence intervals. Each contrast is the product of the DML treatment coefficient and the average rewrite-induced change in the principal component, so its sampling uncertainty has two sources, the cluster bootstrap over the rewritten listings and the estimation error in the DML coefficient itself. The per-market intervals we report fold both into a single standard error by the delta method, adding the coefficient's contribution to the within-listing cluster bootstrap rather than conditioning on the fitted coefficient. This widens the intervals materially relative to a bootstrap that treats the coefficient as known, and it is the reason the pooled statistic below is moderate rather than the implausibly large value a coefficient-conditional interval would produce. Appendix I sets out the rewriting protocol, the two generators, and the delta-method variance in full.

Per-market style-stripped contrast. Table 14 reports the per-market style-stripped contrast at $n = 200$ listings each. The contrast is positive and excludes zero in eight markets and negative and excludes zero in one, with three markets covering zero once the coefficient uncertainty is included. The largest positive contrasts are in Boston at +0.184 on log price, an implied +20.2%, and Atlanta at +0.176 (+19.3%), followed by Denver at +0.162 (+17.6%), Dallas at +0.134 (+14.3%), New York at +0.110 (+11.6%), Portland at +0.088 (+9.2%), Seattle at +0.056 (+5.8%), and San Francisco at +0.033 (+3.4%). Chicago returns a negative contrast of −0.038 that excludes zero, while Washington, Philadelphia, and Phoenix return contrasts whose intervals straddle zero once the coefficient uncertainty is folded in, Phoenix because its point estimate is essentially zero and Washington and Philadelphia because their intervals widen past it. The pooled random-effects style-stripped contrast across the twelve markets is +0.074 on log price, with standard error 0.020, a t statistic of 3.66, and a 95% confidence interval of [+0.034, +0.113], so the pooled contrast is positive and detectable. The between-market heterogeneity is large, with $I^2 = 97.5\%$, so we report the random-effects estimate in preference to the fixed-effects estimate and caution that the pooled figure summarizes a dispersed distribution rather than a single generalizable mean.

Table 14: Counterfactual rewriting intervention: per-market style-stripped and submarket-swap contrasts on log listing price at $n = 200$ listings each, with 95% intervals that fold the DML coefficient's estimation error into the within-listing cluster bootstrap by the delta method. The bottom row is the random-effects pooled contrast.

Market	Style-stripped	95% CI	Submarket-swap
Boston	+0.184	[+0.136, +0.232]	−0.012
New York	+0.110	[+0.092, +0.127]	−0.067
San Francisco	+0.033	[+0.016, +0.050]	+0.015
Washington	−0.005	[−0.015, +0.006]	+0.003
Philadelphia	−0.004	[−0.033, +0.026]	+0.045
Chicago	−0.038	[−0.067, −0.008]	+0.025
Seattle	+0.056	[+0.030, +0.082]	−0.005
Denver	+0.162	[+0.136, +0.189]	+0.039
Atlanta	+0.176	[+0.146, +0.207]	+0.001
Portland	+0.088	[+0.062, +0.114]	−0.008
Phoenix	+0.001	[−0.018, +0.020]	−0.000
Dallas	+0.134	[+0.111, +0.157]	−0.007
Pooled (RE)	+0.074	[+0.034, +0.113]	+0.002

Per-market submarket-swap contrast. The submarket-swap contrast is small in absolute magnitude in every market and splits in sign across the panel. It excludes zero in six markets once the coefficient uncertainty is included, three positive and three negative, and covers zero in the remaining six. The largest negative value is in New York at -0.067 on log price, and the largest positive values are in Philadelphia at $+0.045$ and Denver at $+0.039$. Because the market-level contrasts point in opposite directions and are individually small, they largely cancel in aggregation. The pooled random-effects submarket-swap contrast is $+0.002$ on log price, with a 95% confidence interval of $[-0.007, +0.010]$ that includes zero, so the pooled submarket-swap contrast is statistically indistinguishable from zero even though several markets carry significant contrasts of either sign.

Interpretation. The two contrasts measure different manipulations and need not agree. The positive pooled style-stripped contrast says that, holding the model's inferred location fixed, the valuation model attaches a positive price to the listing's own stylistic and semantic content, on the order of seven to eight percent on average and considerably larger in several markets. The near-zero pooled submarket-swap contrast says that when the rewrite is allowed to move the model's inferred location as well, the net change in predicted price averages to approximately nothing, because the markets divide in sign. We do not read the gap between the two as a formal indirect effect through a location mediator, because the design does not identify one, but the pattern is consistent with the dual role of listing text documented by Shen and Ross [2021] and Baur et al. [2023a], in which descriptive language carries both an own-contribution to valuation and information that the model reads as a proxy for location. We treat the two contrasts as auditable probes of how one deployed valuation model responds to controlled edits of the input text, not as structural estimates of a causal mechanism.

Cross-generator robustness. We re-ran the rewriting intervention with a second open-weight generator, Llama 3.3 70B Instruct, to gauge how far the conclusions depend on the choice of rewriting model, and we compare it against the primary Qwen 2.5 32B Instruct run. The two runs are not a matched replication. The Llama run processes $n = 100$ listings per market against the $n = 200$ of the primary run, so the comparison is an unmatched cross-generator probe of direction and rough magnitude rather than a tight correlation between estimates computed on identical samples. Across the twelve markets the per-market style-stripped contrasts from the two generators agree in sign in nine of the twelve markets, with a Spearman rank correlation of 0.46 and a Pearson correlation of 0.23 on the point estimates. The submarket-swap contrasts agree in sign in six of the twelve markets, with a Spearman correlation of 0.64 and a Pearson correlation of 0.82. The central tendencies move in the same direction across generators for the style-stripped contrast, with a positive average in both runs, the random-effects pooled $+0.074$ under Qwen and a per-market mean of $+0.070$ under Llama. The submarket-swap contrast is the less stable of the two across generators, near zero under Qwen and modestly negative under Llama, which is unsurprising given that it is the small residual after two larger channels nearly cancel. We read the cross-generator evidence as directional support for a positive style-stripped contrast and a near-zero to slightly negative submarket-swap contrast, and we are explicit that the rank and linear correlations are moderate rather than near-perfect.